Set 2C: Op-Amp Filter Design from a Bode Plot

ECEN 325 — Lecture 2 (Operational Amplifiers) practice set

What this set trains. A design problem runs the pipeline backwards: read $H(s)$ from a straight-line Bode plot (exactly as in Lecture 1, set 1D), split it into first-order factors, realize each group of factors with one inverting op-amp stage, and compute the component values. Design rules used in every problem (they make the answer unique so you can check it): $R_1 = 10\,\text{k}\Omega$ at the input of every stage; all of $|K|$ in the first stage; a unity-gain inverter at the end if needed so that $K > 0$. Then find the steady-state output for the given input. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-2 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1 tools for impedances, dividers, Bode plots and sinusoids):

T1 *Ideal op-amp with negative feedback:* no current into + or −; $V_- = V_+$ (virtual short). Its output is an ideal voltage source (loads on it do not matter).

T2 *Block formulas:* inverting $\frac{V_o}{V_{in}} = -\frac{Z_f}{Z_{in}}$; non-inverting $\frac{V_o}{V_+} = 1 + \frac{Z_f}{Z_g}$ (with V_+ from the divider at the + input); a follower has gain 1. Several blocks in cascade: multiply.

T3 *KCL with impedances:* at a node, $\sum_k \frac{V_{node} - V_k}{Z_k} = 0$ (for a capacitor $\frac{1}{Z} = sC$). Write it at every node except op-amp outputs and source nodes.

T4 *Solving KCL equations:* pick the equation with the fewest unknowns, solve it for one node voltage, substitute into the rest; repeat until only V_o and V_i remain.

T5 *Finite gain:* $V_{out} = A(s)(V_+ - V_-)$, $A(s) = \frac{A_0}{1 + s/\omega_b}$, $\omega_t = A_0\omega_b$. With $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$: $\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta}$.

T6 *Slew rate:* for $V_{out} = \hat{V}\sin\omega t$, the fastest change is $\left|\frac{dV_{out}}{dt}\right|_{max} = \omega\hat{V}$; no distortion requires $\omega\hat{V} \le SR$.

Steps:

1. *Sources.* DC and AC both present ⇒ superposition.
2. *DC circuit:* $V_i \to 0$, $C \to$ open; pass the DC through each block with its gain at $s = 0$.
3. *Op-amp rules:* confirm negative feedback; apply T1 (or T5 for a non-ideal op-amp).
4. *Transfer function:* standard blocks → T2; anything else → T3 + T4.
5. *Standard form*, 6. *Bode*, 7. *Output* — exactly as in Lecture 1.

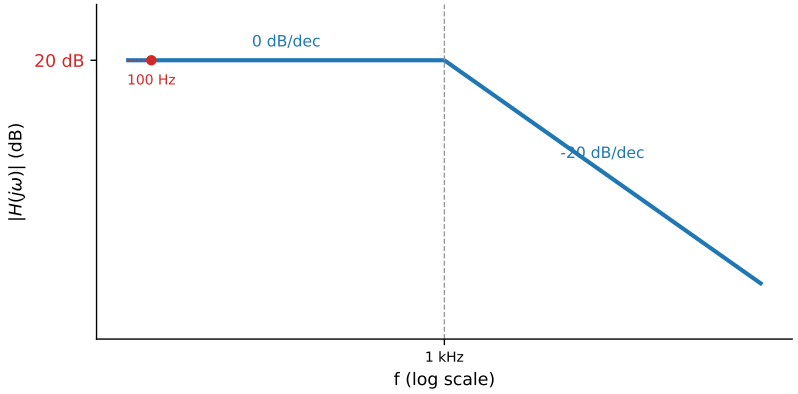
Variations: a design problem runs these steps backwards (Bode plot → $H(s)$ → blocks → component values); a slew-rate problem stops after Step 4 for every op-amp output and applies T6.

Problems

Easy problems

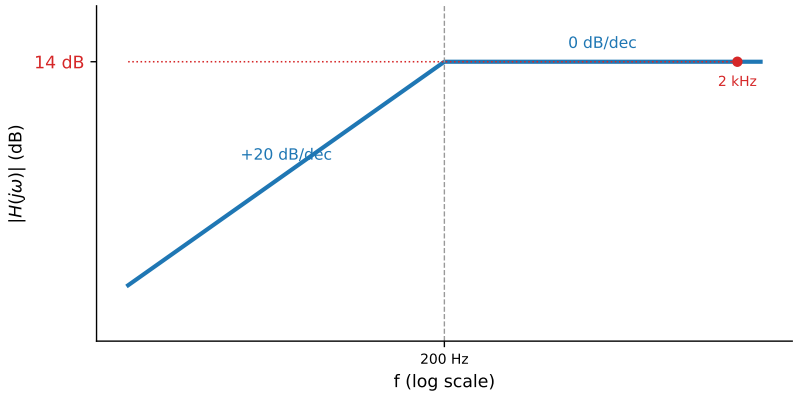
Problem 2C.1 — Easy

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 2000\,t)$.



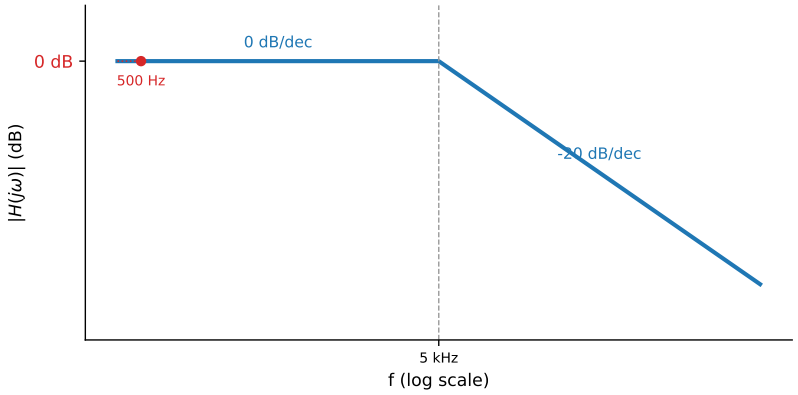
Problem 2C.2 — Easy

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 100\,t)$.



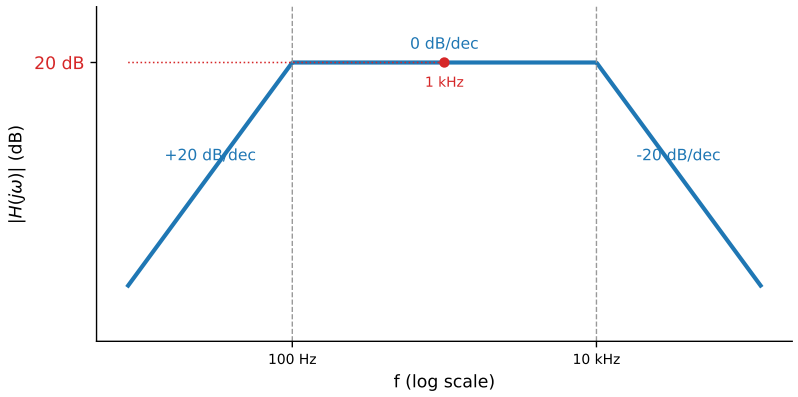
Problem 2C.3 — Easy

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 1\cos(2\pi \cdot 10000\,t)$.



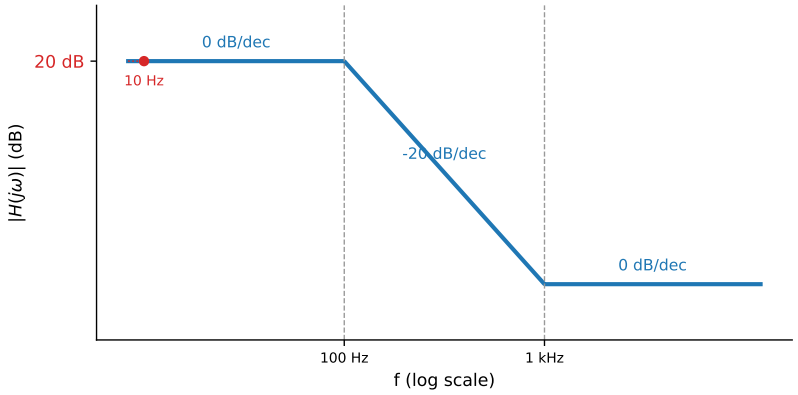
Problem 2C.4 — Easy

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 300\,t)$.



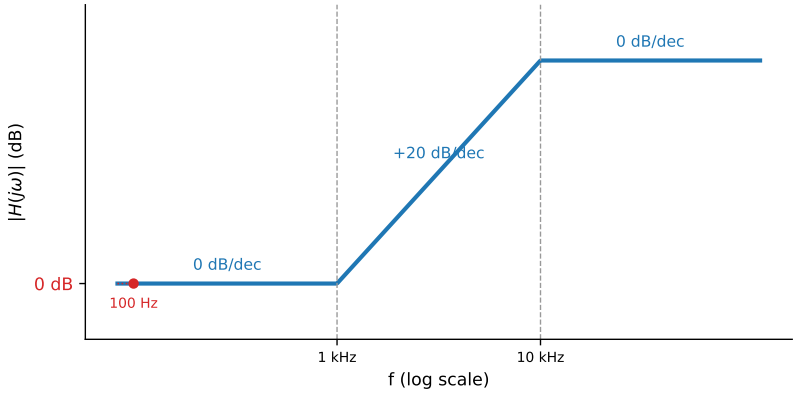
Problem 2C.5 — Easy

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 500\,t)$.



Problem 2C.6 — Easy

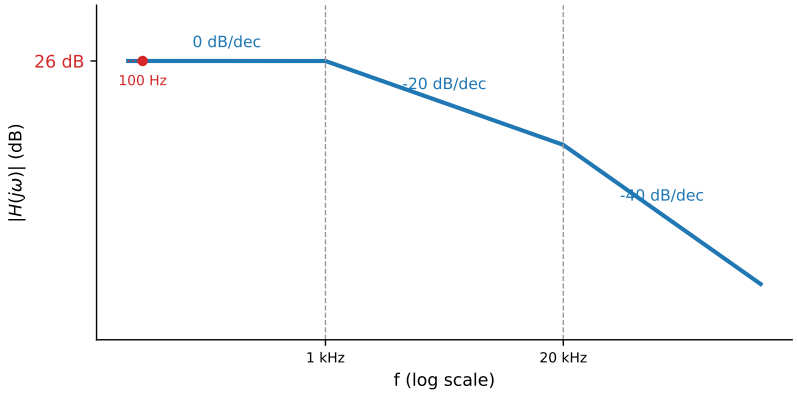
Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 3000\,t)$.



Medium problems

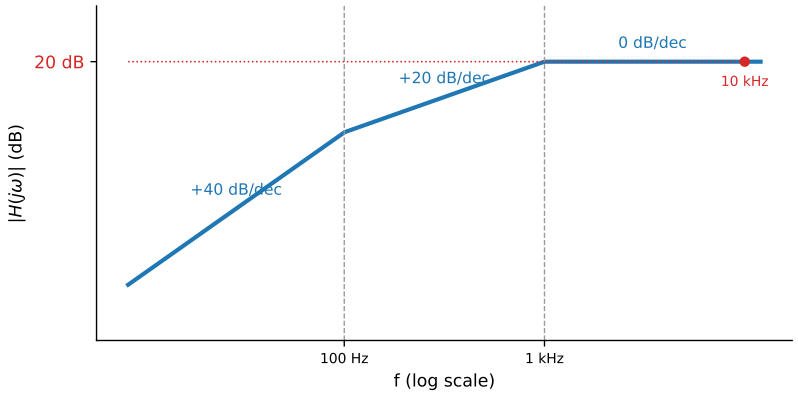
Problem 2C.7 — Medium

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 5000\,t)$.



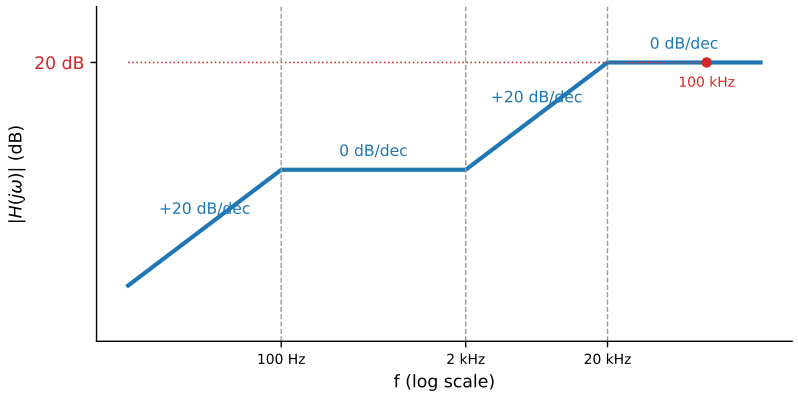
Problem 2C.8 — Medium

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 300\,t)$.



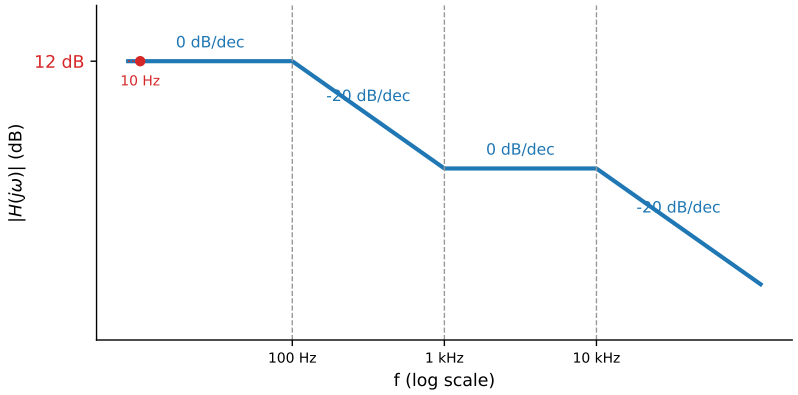
Problem 2C.9 — Medium

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 1000\,t)$.



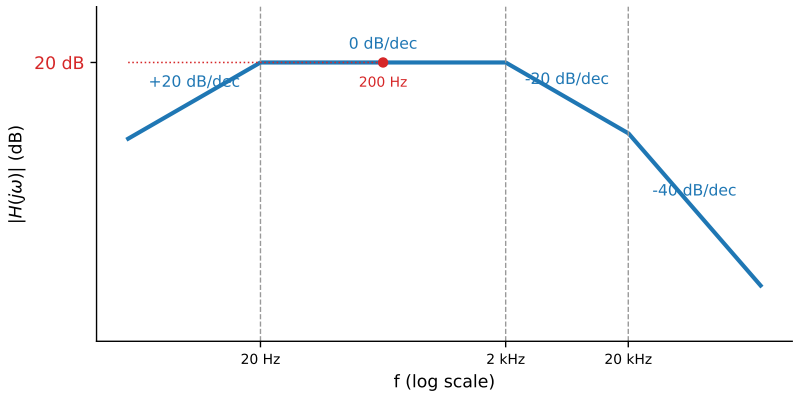
Problem 2C.10 — Medium

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 2000\,t)$.



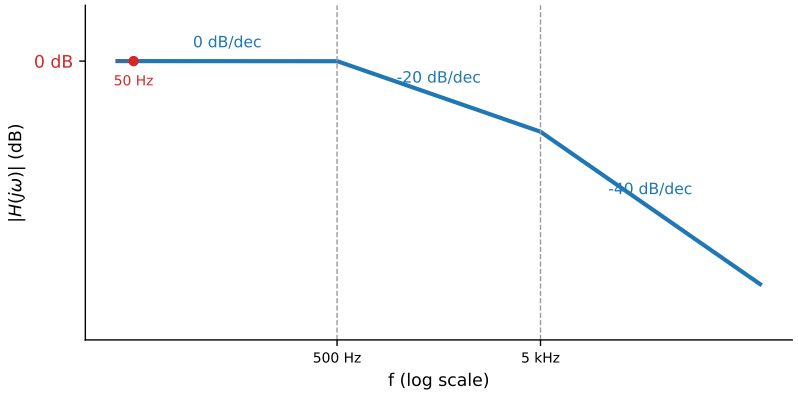
Problem 2C.11 — Medium

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.1 \sin(2\pi \cdot 5000\, t)$.



Problem 2C.12 — Medium

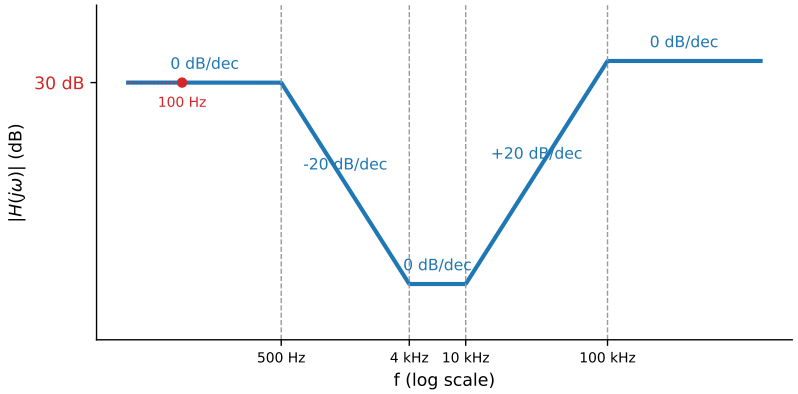
Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 1 \sin\left(2\pi \cdot 1000\, t + \frac{\pi}{4}\right)$.



Hard problems

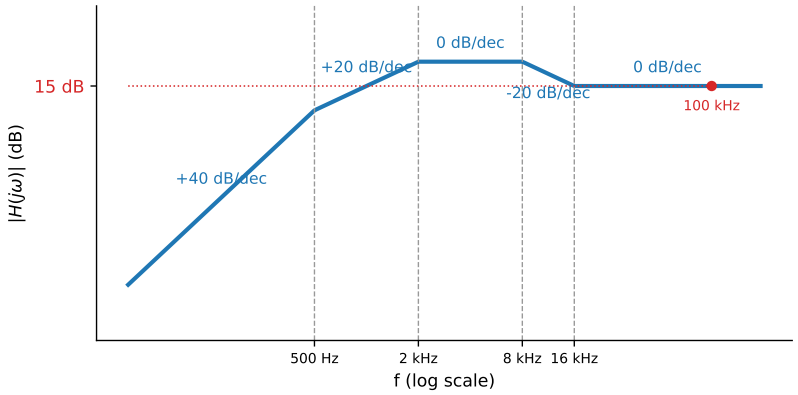
Problem 2C.13 — Hard

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.01 \sin(2\pi \cdot 2000\, t)$.



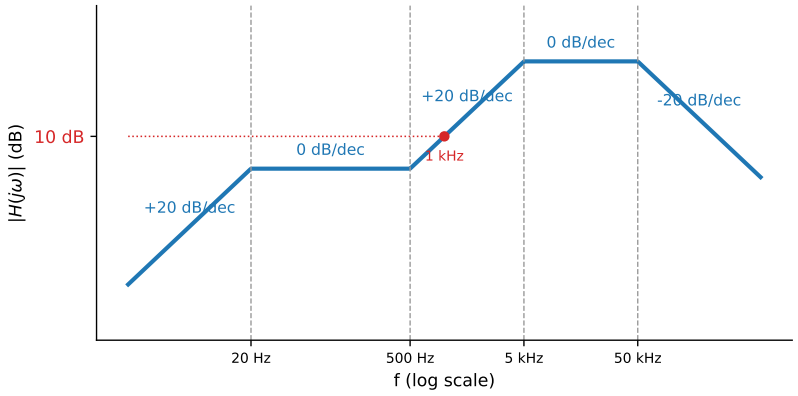
Problem 2C.14 — Hard

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.3 \sin(2\pi \cdot 10000\, t)$.



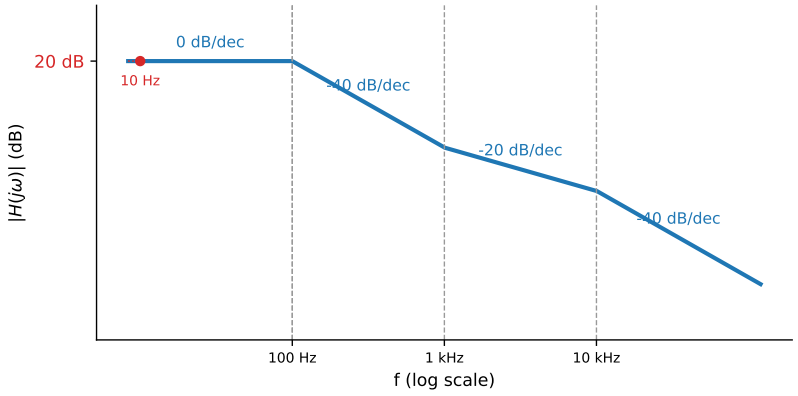
Problem 2C.15 — Hard

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin\left(2\pi \cdot 2000\,t + \frac{\pi}{3}\right)$.



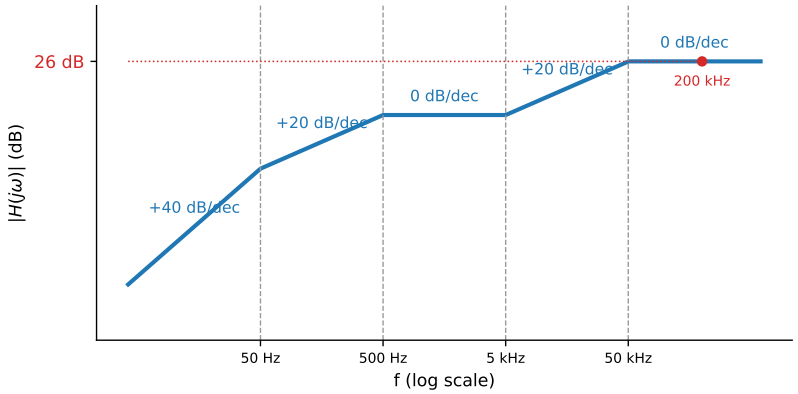
Problem 2C.16 — Hard

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin\left(2\pi \cdot 300\,t\right)$.



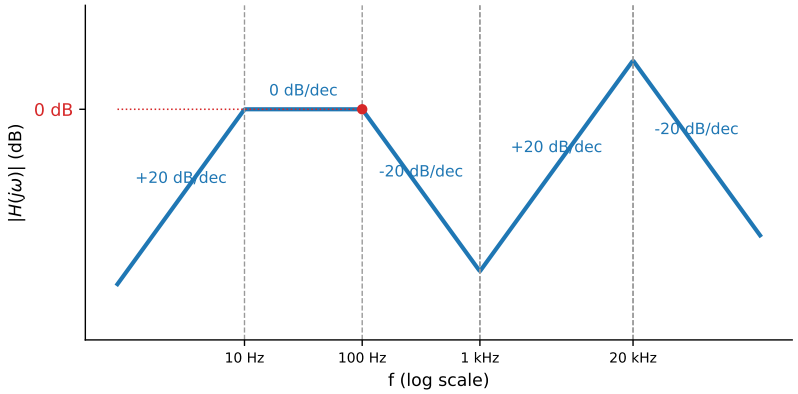
Problem 2C.17 — Hard

Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\cos\left(2\pi \cdot 1000\,t - \frac{\pi}{6}\right)$.



Problem 2C.18 — Hard

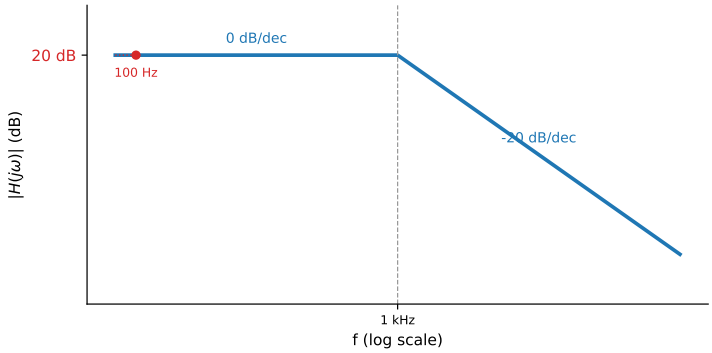
Design a circuit with op-amps, resistors and capacitors whose transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ has the straight-line magnitude Bode plot below ($K > 0$, no right-half-plane poles or zeros; use the design rules stated at the top of this set). Then find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin\left(2\pi \cdot 3000\,t\right)$.



Solutions

Every solution follows the steps of the Lecture-2 pipeline card.

Solution 2C.1 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
1 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 20 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}$	1 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 20 - (0) = 20\text{ dB} \quad\Rightarrow\quad K = 10^{20/20} = 10$$

So

$$\boxed{H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	10	$\dfrac{1}{1+\frac{s}{2\pi\cdot 1\text{k}}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$
U2	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

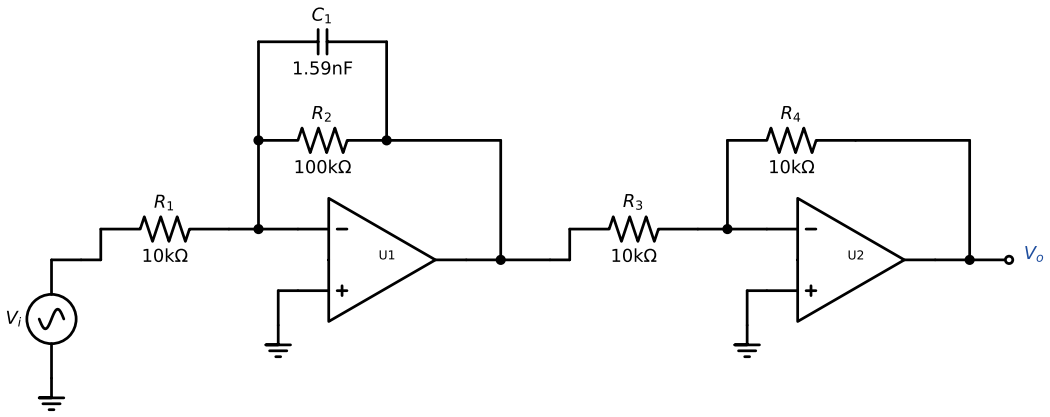
1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{1}{1+sR_2C_1}$ to the factors:

$$R_2 = 10\,R_1 = 100\,\text{k}\Omega, \qquad C_1 = \frac{1}{6280\,R_2} = 1.59\,\text{nF}$$

2. U2: $R_3 = R_4 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 10 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.1\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	10	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1\text{k}}}$	0.4472	-1.11
total	**4.472** (product)	$\mathbf{-1.11}$ (sum)

$$|H(j\omega_{in})| = 4.47, \qquad \angle H(j\omega_{in}) = -1.11 \text{ rad}$$
$$v_{o,ac}(t) = 4.47 \times 0.1 \sin(2\pi \cdot 2000\,t - 1.11) = 0.447 \sin(2\pi \cdot 2000\,t - 1.11)$$

$$\boxed{V_o(t) = 0.447 \sin(2\pi \cdot 2000\,t - 1.11)\ \text{V}}$$

Answer

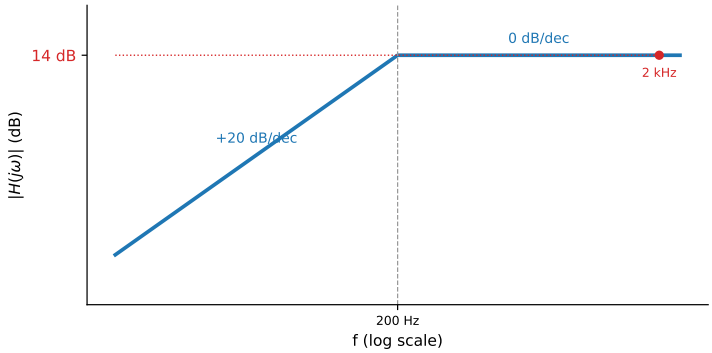
$$H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi \cdot 1\text{k}}}$$

U1: R1 = 10kΩ, R2 = 100kΩ, C1 = 1.59nF

U2: R3 = 10kΩ, R4 = 10kΩ

$V_o(t) = 0.447 \sin(2\pi \cdot 2000\,t - 1.11)$ V

Solution 2C.2 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
200 Hz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (200 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s + 2\pi\cdot 200}$$

Step 4 — Find K from the marked level

The plot marks the level 14 dB at $f = 2$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 2$ kHz	asymptote (dB)
$\dfrac{s}{s+2\pi\cdot 200}$	200 Hz	$f > f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 14 - (0) = 14 \text{ dB} \quad \Rightarrow \quad K = 10^{14/20} = 5.01$$

So

$$\boxed{H(s) = 5.01\,\frac{s}{s + 2\pi \cdot 200}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	5.012	$\dfrac{s}{s + 2\pi \cdot 200}$	high-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R$
U2	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

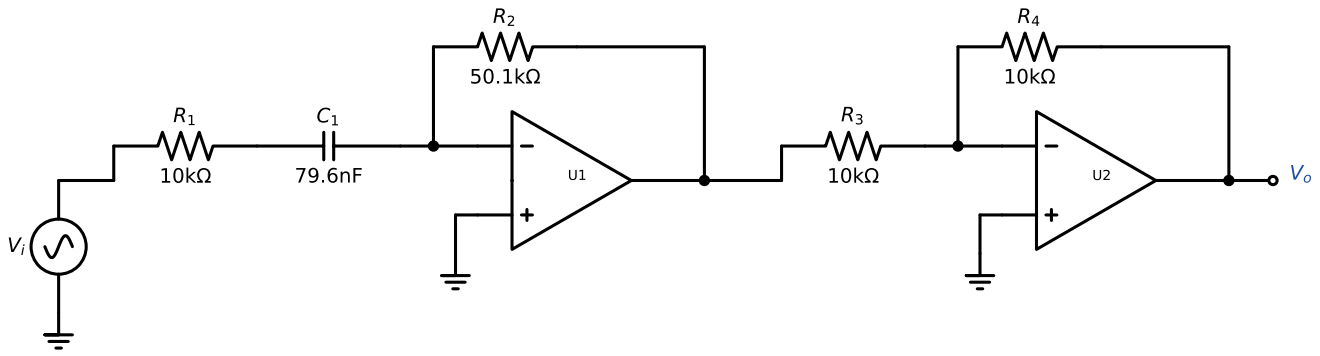
1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1} \cdot \frac{s}{s+1/(R_1C_1)}$ to the factors:

$$C_1 = \frac{1}{1260\,R_1} = 79.6\,\text{nF}, \qquad R_2 = 5.012\,R_1 = 50.1\,\text{k}\Omega$$

2. U2: $R_3 = R_4 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 5.012 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.2\sin(2\pi \cdot 100\,t)$, so $\omega_{in} = 2\pi \cdot 100 = 628\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1 + \frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
5.01	5.012	0
$\dfrac{s}{s+2\pi\cdot 200}$	0.4472	1.11
total	**2.241** (product)	**1.11** (sum)

$$|H(j\omega_{in})| = 2.24, \qquad \angle H(j\omega_{in}) = 1.11 \text{ rad}$$
$$v_{o,ac}(t) = 2.24 \times 0.2 \sin(2\pi\cdot 100\,t + 1.11) = 0.448 \sin(2\pi\cdot 100\,t + 1.11)$$

$$\boxed{V_o(t) = 0.448 \sin(2\pi\cdot 100\,t + 1.11) \text{ V}}$$

Answer

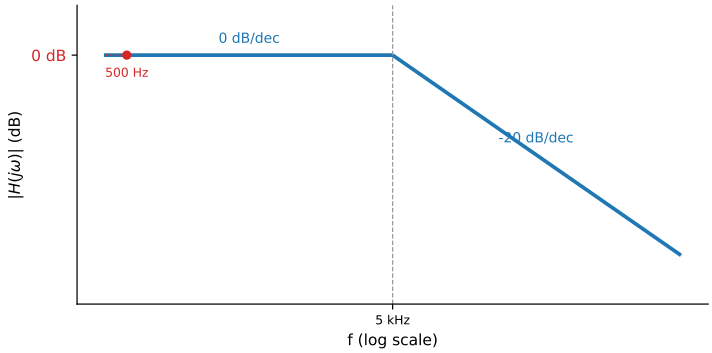
$H(s) = 5.01 \dfrac{s}{s+2\pi\cdot 200}$

U1: R1 = 10kΩ, C1 = 79.6nF, R2 = 50.1kΩ

U2: R3 = 10kΩ, R4 = 10kΩ

$V_o(t) = 0.448 \sin(2\pi\cdot 100\,t + 1.11)$ V

Solution 2C.3 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
5 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 0 dB at $f = 500$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 500$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}$	5 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 0 - (0) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	1	$\dfrac{1}{1+\frac{s}{2\pi\cdot 5\text{k}}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$
U2	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

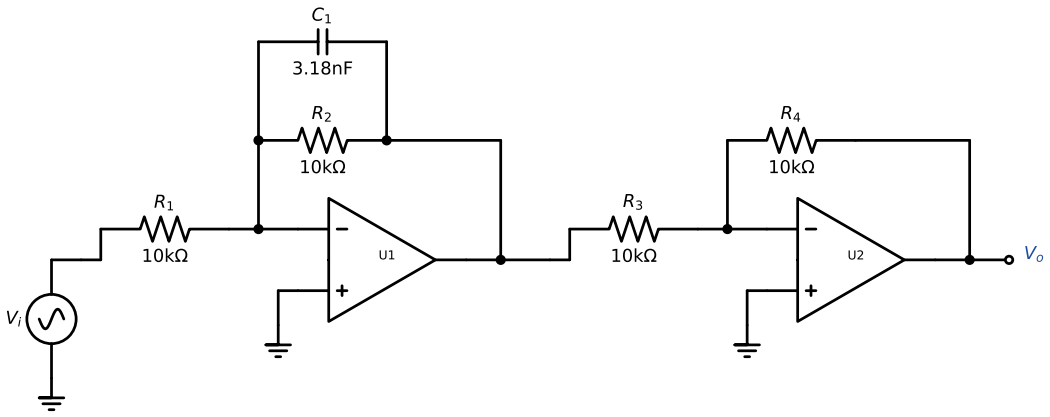
1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{1}{1+sR_2C_1}$ to the factors:

$$R_2 = 1\,R_1 = 10\,\text{k}\Omega, \qquad C_1 = \frac{1}{31400\,R_2} = 3.18\,\text{nF}$$

2. U2: $R_3 = R_4 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 1 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 1\cos(2\pi \cdot 10000\,t)$, so $\omega_{in} = 2\pi \cdot 10000 = 62800$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 5\text{k}}}$	0.4472	-1.11
total	**0.4472** (product)	$\mathbf{-1.11}$ (sum)

$$|H(j\omega_{in})| = 0.447, \qquad \angle H(j\omega_{in}) = -1.11 \text{ rad}$$
$$v_{o,ac}(t) = 0.447 \times 1\cos(2\pi \cdot 10000\,t - 1.11) = 0.447\cos(2\pi \cdot 10000\,t - 1.11)$$

$$\boxed{V_o(t) = 0.447\cos(2\pi \cdot 10000\,t - 1.11)\ \text{V}}$$

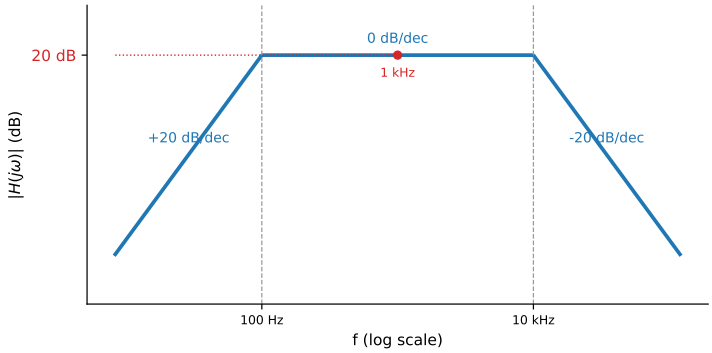
Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi \cdot 5\text{k}}}$$

U1: R1 = 10kΩ, R2 = 10kΩ, C1 = 3.18nF
U2: R3 = 10kΩ, R4 = 10kΩ

$$V_o(t) = 0.447\cos(2\pi \cdot 10000\,t - 1.11)\ \text{V}$$

Solution 2C.4 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	$+20$	0	-20	one pole at f_c
10 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (100 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 100}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 20 dB at $f = 1$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 1$ kHz	asymptote (dB)
$\dfrac{s}{s+2\pi\cdot 100}$	100 Hz	$f > f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	10 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 20 - (0) = 20\ \text{dB} \quad \Rightarrow \quad K = 10^{20/20} = 10$$

So

$$\boxed{H(s) = 10\,\frac{s}{s + 2\pi\cdot 100}\,\frac{1}{1 + \dfrac{s}{2\pi\cdot 10\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	10	$\dfrac{s}{s + 2\pi\cdot 100}\cdot\dfrac{1}{1 + \frac{s}{2\pi\cdot 10\text{k}}}$	band-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R \parallel C$
U2	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

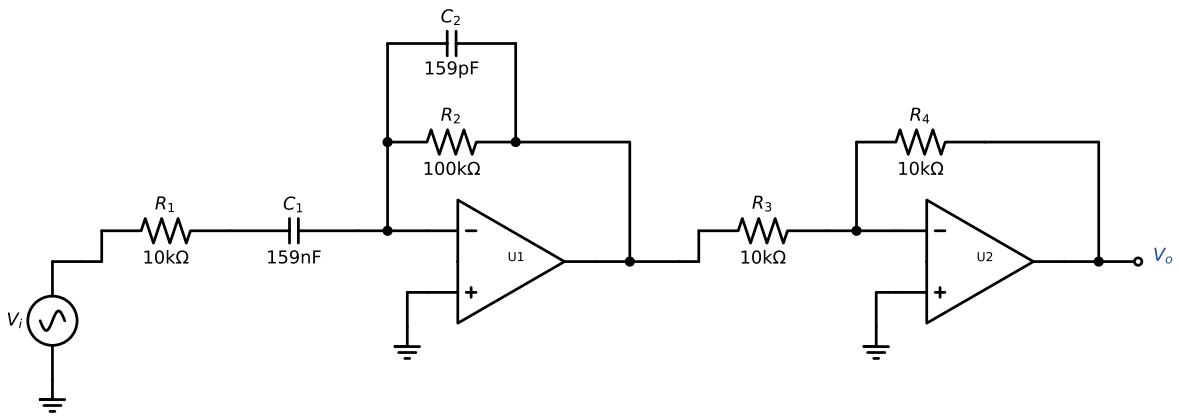
1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{s}{s+1/(R_1C_1)}\cdot\frac{1}{1+sR_2C_2}$ to the factors:

$$C_1 = \frac{1}{628\,R_1} = 159\,\text{nF}, \qquad R_2 = 10\,R_1 = 100\,\text{k}\Omega, \qquad C_2 = \frac{1}{62800\,R_2} = 159\,\text{pF}$$

2. U2: $R_3 = R_4 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 10 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.05\sin(2\pi \cdot 300\,t)$, so $\omega_{in} = 2\pi \cdot 300 = 1880\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	10	0
$\dfrac{s}{s+2\pi\cdot 100}$	0.9487	0.322
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	0.9996	−0.03
total	**9.483** (product)	**0.292** (sum)

$$|H(j\omega_{in})| = 9.48, \qquad \angle H(j\omega_{in}) = 0.292 \text{ rad}$$

$$v_{o,ac}(t) = 9.48 \times 0.05\,\sin(2\pi \cdot 300\,t + 0.292) = 0.474\,\sin(2\pi \cdot 300\,t + 0.292)$$

$$\boxed{V_o(t) = 0.474\,\sin(2\pi \cdot 300\,t + 0.292)\ \text{V}}$$

Answer

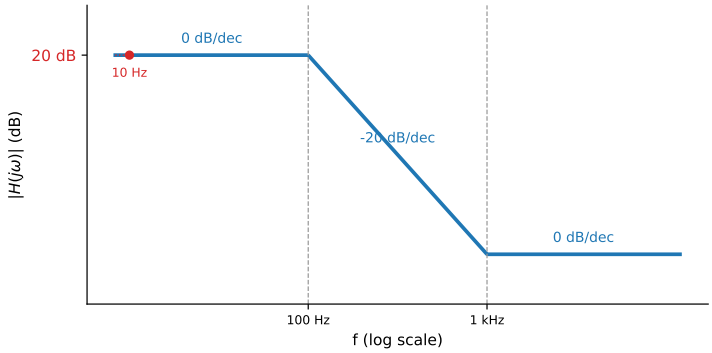
$$H(s) = 10\,\frac{s}{s+2\pi\cdot 100}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$$

U1: R1 = 10kΩ, C1 = 159nF, R2 = 100kΩ, C2 = 159pF

U2: R3 = 10kΩ, R4 = 10kΩ

$V_o(t) = 0.474\,\sin(2\pi \cdot 300\,t + 0.292)$ V

Solution 2C.5 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	0	-20	-20	one pole at f_c
1 kHz	-20	0	$+20$	one zero at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)$$

Step 4 — Find K from the marked level

The plot marks the level 20 dB at $f = 10$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 10$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	100 Hz	$f < f_c$: ≈ 1	0
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 20 - (0) = 20 \text{ dB} \quad \Rightarrow \quad K = 10^{20/20} = 10$$

So

$$\boxed{H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	10	$\dfrac{1+\frac{s}{2\pi\cdot 1\text{k}}}{1+\frac{s}{2\pi\cdot 100}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$
U2	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

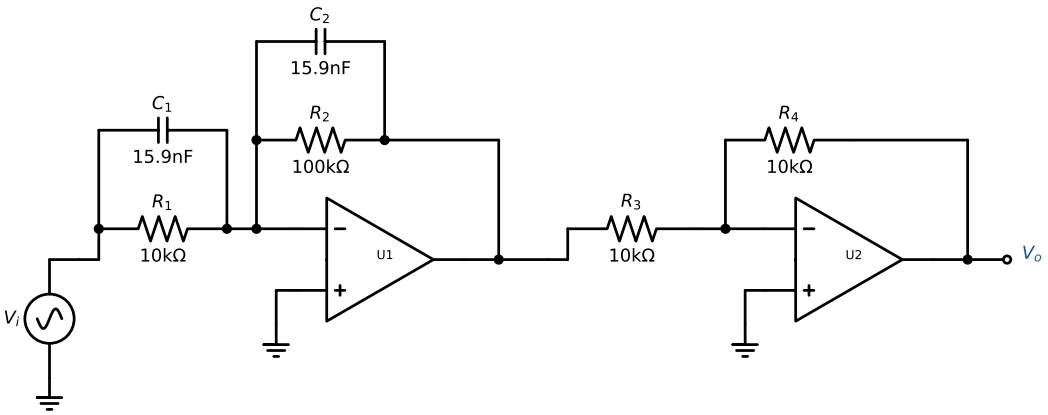
1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{1+sR_1C_1}{1+sR_2C_2}$ to the factors:

$$R_2 = 10\,R_1 = 100\,\text{k}\Omega, \qquad C_1 = \frac{1}{6280\,R_1} = 15.9\,\text{nF}, \qquad C_2 = \frac{1}{628\,R_2} = 15.9\,\text{nF}$$

2. U2: $R_3 = R_4 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 10 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.1 \sin(2\pi \cdot 500\, t)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	10	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 100}}$	0.1961	-1.37
$\left(1+\dfrac{s}{2\pi \cdot 1\text{k}}\right)$	1.118	0.464
total	**2.193** (product)	$-$**0.91** (sum)

$$|H(j\omega_{in})| = 2.19, \qquad \angle H(j\omega_{in}) = -0.91 \text{ rad}$$
$$v_{o,ac}(t) = 2.19 \times 0.1 \sin(2\pi \cdot 500\, t - 0.91) = 0.219 \sin(2\pi \cdot 500\, t - 0.91)$$

$$\boxed{V_o(t) = 0.219 \sin(2\pi \cdot 500\, t - 0.91) \text{ V}}$$

Answer

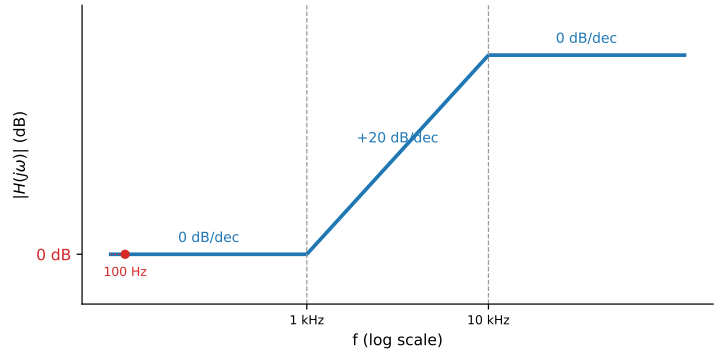
$$H(s) = 10 \frac{1}{1+\dfrac{s}{2\pi \cdot 100}} \left(1+\frac{s}{2\pi \cdot 1\text{k}}\right)$$

U1: R1 = 10kΩ, C1 = 15.9nF, R2 = 100kΩ, C2 = 15.9nF

U2: R3 = 10kΩ, R4 = 10kΩ

$V_o(t) = 0.219 \sin(2\pi \cdot 500\, t - 0.91)$ V

Solution 2C.6 **Easy**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
1 kHz	0	$+20$	$+20$	one zero at f_c
10 kHz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 0 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	10 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 0 - (0) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1\left(1 + \frac{s}{2\pi \cdot 1\text{k}}\right)\frac{1}{1 + \frac{s}{2\pi \cdot 10\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	1	$\frac{1 + \frac{s}{2\pi\cdot 1\text{k}}}{1 + \frac{s}{2\pi\cdot 10\text{k}}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$
U2	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

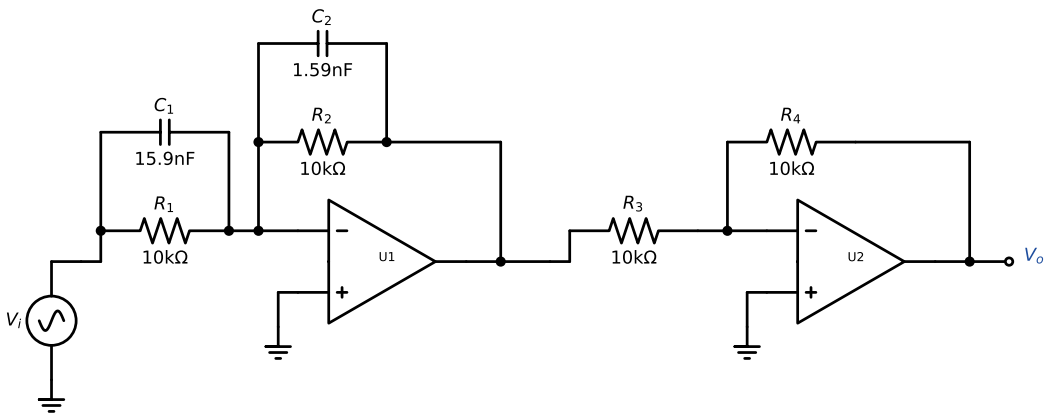
1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1} \cdot \frac{1+sR_1C_1}{1+sR_2C_2}$ to the factors:

$$R_2 = 1\,R_1 = 10\,\text{k}\Omega, \qquad C_1 = \frac{1}{6280\,R_1} = 15.9\,\text{nF}, \qquad C_2 = \frac{1}{62800\,R_2} = 1.59\,\text{nF}$$

2. U2: $R_3 = R_4 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 1 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.5\sin(2\pi \cdot 3000\,t)$, so $\omega_{in} = 2\pi \cdot 3000 = 18800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\left(1+\dfrac{s}{2\pi \cdot 1\text{k}}\right)$	3.162	1.25
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 10\text{k}}}$	0.9578	-0.291
total	**3.029** (product)	**0.958** (sum)

$$|H(j\omega_{in})| = 3.03, \qquad \angle H(j\omega_{in}) = 0.958 \text{ rad}$$
$$v_{o,ac}(t) = 3.03 \times 0.5 \sin(2\pi \cdot 3000\,t + 0.958) = 1.51 \sin(2\pi \cdot 3000\,t + 0.958)$$

$$\boxed{V_o(t) = 1.51 \sin(2\pi \cdot 3000\,t + 0.958)\ \text{V}}$$

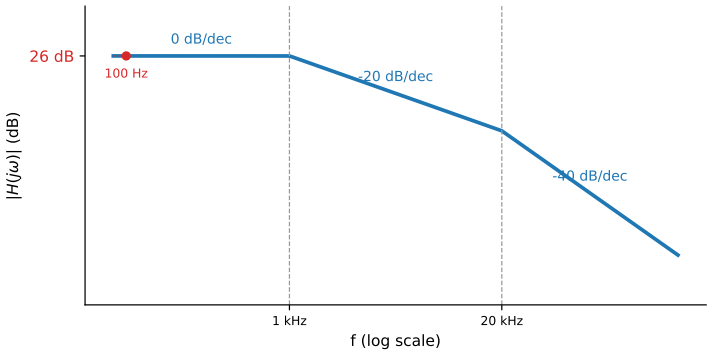
Answer

$$H(s) = 1\left(1+\frac{s}{2\pi \cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi \cdot 10\text{k}}}$$

U1: R1 = 10kΩ, C1 = 15.9nF, R2 = 10kΩ, C2 = 1.59nF
U2: R3 = 10kΩ, R4 = 10kΩ

$V_o(t) = 1.51 \sin(2\pi \cdot 3000\,t + 0.958)$ V

Solution 2C.7 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
1 kHz	0	-20	-20	one pole at f_c
20 kHz	-20	-40	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 26 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}$	1 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	20 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 26 - (0) = 26 \text{ dB} \quad \Rightarrow \quad K = 10^{26/20} = 20$$

So

$$\boxed{H(s) = 20\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	19.95	$\dfrac{1}{1+\frac{s}{2\pi\cdot 1\text{k}}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$
U2	1	$\dfrac{1}{1+\frac{s}{2\pi\cdot 20\text{k}}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{1}{1+sR_2C_1}$ to the factors:

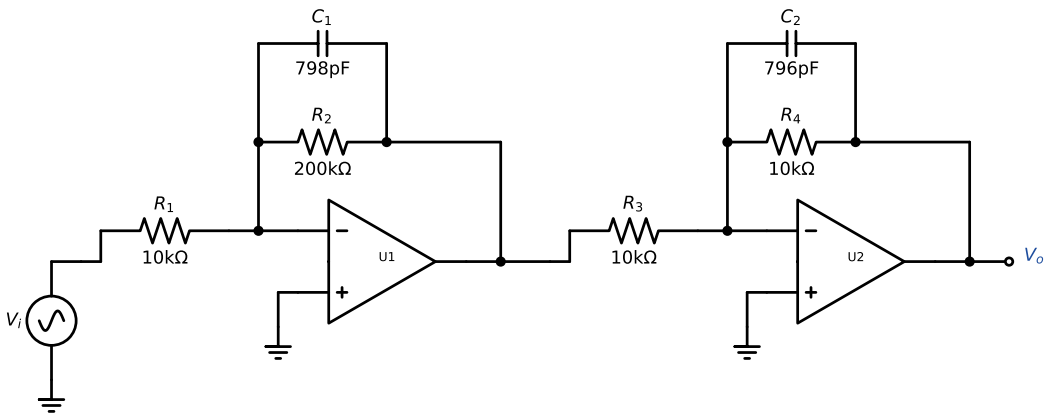
$$R_2 = 19.95\,R_1 = 200\,\text{k}\Omega, \qquad C_1 = \frac{1}{6280\,R_2} = 798\,\text{pF}$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3}\cdot\frac{1}{1+sR_4C_2}$ to the factors:

$$R_4 = 1\,R_3 = 10\,\text{k}\Omega, \qquad C_2 = \frac{1}{1.26\times 10^5\,R_4} = 796\,\text{pF}$$

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 19.95 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.05 \sin(2\pi \cdot 5000\, t)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
20	19.95	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1\text{k}}}$	0.1961	-1.37
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 20\text{k}}}$	0.9701	-0.245
total	**3.796** (product)	$-$**1.62** (sum)

$$|H(j\omega_{in})| = 3.8, \qquad \angle H(j\omega_{in}) = -1.62 \text{ rad}$$
$$v_{o,ac}(t) = 3.8 \times 0.05 \sin(2\pi \cdot 5000\, t - 1.62) = 0.19 \sin(2\pi \cdot 5000\, t - 1.62)$$

$$\boxed{V_o(t) = 0.19 \sin(2\pi \cdot 5000\, t - 1.62) \text{ V}}$$

Answer

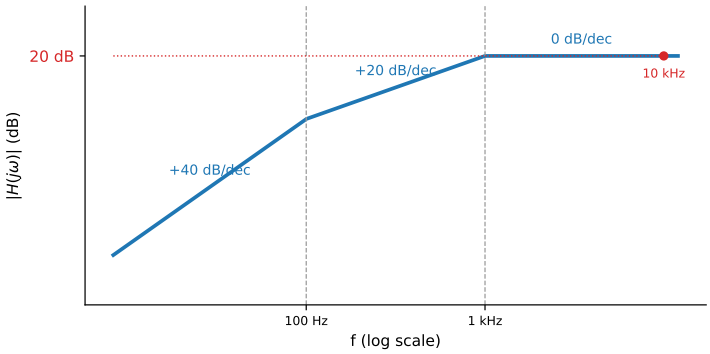
$$H(s) = 20 \frac{1}{1+\dfrac{s}{2\pi \cdot 1\text{k}}} \frac{1}{1+\dfrac{s}{2\pi \cdot 20\text{k}}}$$

U1: R1 = 10kΩ, R2 = 200kΩ, C1 = 798pF

U2: R3 = 10kΩ, R4 = 10kΩ, C2 = 796pF

$V_o(t) = 0.19 \sin(2\pi \cdot 5000\, t - 1.62)$ V

Solution 2C.8 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	$+40$	$+20$	-20	one pole at f_c
1 kHz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+40$ dB/dec, i.e. $|H| \propto \omega^2$ at low frequency: the numerator contains s^2. As in class, pair each s with one of the *lowest* poles (100 Hz, 1 kHz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 100}\,\frac{s}{s+2\pi\cdot 1\text{k}}$$

Step 4 — Find K from the marked level

The plot marks the level 20 dB at $f = 10$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 10$ kHz	asymptote (dB)
$\frac{s}{s+2\pi\cdot 100}$	100 Hz	$f > f_c$: ≈ 1	0
$\frac{s}{s+2\pi\cdot 1\text{k}}$	1 kHz	$f > f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10}K = 20 - (0) = 20\ \text{dB} \quad \Rightarrow \quad K = 10^{20/20} = 10$$

So

$$\boxed{H(s) = 10\,\frac{s}{s+2\pi\cdot 100}\,\frac{s}{s+2\pi\cdot 1\text{k}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	10	$\dfrac{s}{s+2\pi\cdot 100}$	high-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R$
U2	1	$\dfrac{s}{s+2\pi\cdot 1\text{k}}$	high-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R$

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{s}{s+1/(R_1C_1)}$ to the factors:

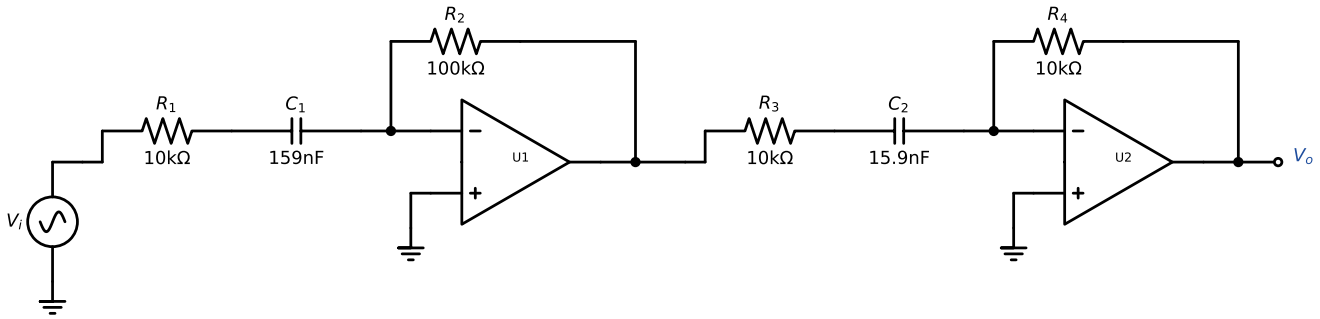
$$C_1 = \frac{1}{628\,R_1} = 159\,\text{nF}, \qquad R_2 = 10\,R_1 = 100\,\text{k}\Omega$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3}\cdot\frac{s}{s+1/(R_3C_2)}$ to the factors:

$$C_2 = \frac{1}{6280\,R_3} = 15.9\,\text{nF}, \qquad R_4 = 1\,R_3 = 10\,\text{k}\Omega$$

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 10 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.1 \sin(2\pi \cdot 300\, t)$, so $\omega_{in} = 2\pi \cdot 300 = 1880\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	10	0
$\dfrac{s}{s+2\pi\cdot 100}$	0.9487	0.322
$\dfrac{s}{s+2\pi\cdot 1\text{k}}$	0.2873	1.28
total	**2.726** (product)	**1.6** (sum)

$$|H(j\omega_{in})| = 2.73, \qquad \angle H(j\omega_{in}) = 1.6 \text{ rad}$$
$$v_{o,ac}(t) = 2.73 \times 0.1 \sin(2\pi \cdot 300\, t + 1.6) = 0.273 \sin(2\pi \cdot 300\, t + 1.6)$$

$$\boxed{V_o(t) = 0.273 \sin(2\pi \cdot 300\, t + 1.6) \text{ V}}$$

Answer

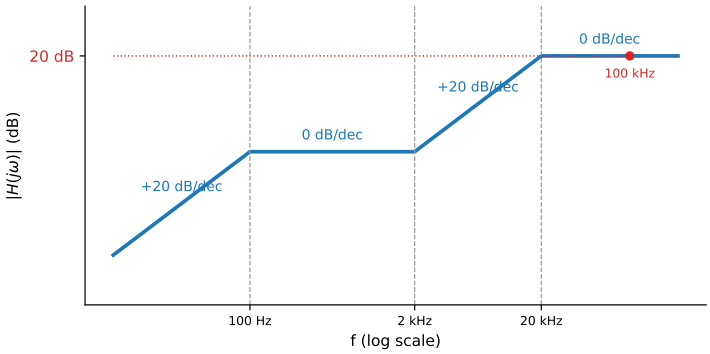
$H(s) = 10 \dfrac{s}{s+2\pi\cdot 100} \dfrac{s}{s+2\pi\cdot 1\text{k}}$

U1: R1 = 10kΩ, C1 = 159nF, R2 = 100kΩ

U2: R3 = 10kΩ, C2 = 15.9nF, R4 = 10kΩ

$V_o(t) = 0.273 \sin(2\pi \cdot 300\, t + 1.6)$ V

Solution 2C.9 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	$+20$	0	-20	one pole at f_c
2 kHz	0	$+20$	$+20$	one zero at f_c
20 kHz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope $\Rightarrow$ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (100 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 100}\left(1+\frac{s}{2\pi\cdot 2\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 20 dB at $f = 100$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ kHz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 100}$	100 Hz	$f > f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 2\text{k}}\right)$	2 kHz	$f > f_c$: $\approx \frac{\omega}{z}$	34
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}$	20 kHz	$f > f_c$: $\approx \frac{p}{\omega}$	-14
sum			**20**

The dB values of the factors *add* (magnitudes multiply), so

$$20 \log_{10} K = 20 - (20) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1 \, \frac{s}{s + 2\pi \cdot 100} \left(1 + \frac{s}{2\pi \cdot 2\text{k}}\right) \frac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	1	$\dfrac{s}{s + 2\pi \cdot 100}$	high-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R$
U2	1	$\dfrac{1 + \frac{s}{2\pi \cdot 2\text{k}}}{1 + \frac{s}{2\pi \cdot 20\text{k}}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1} \cdot \frac{s}{s+1/(R_1 C_1)}$ to the factors:

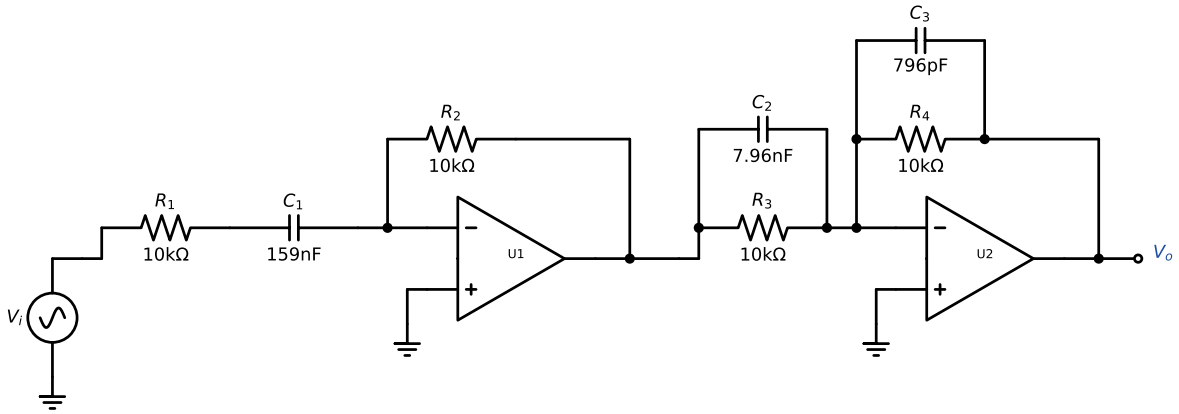
$$C_1 = \frac{1}{628\, R_1} = 159\,\text{nF}, \qquad R_2 = 1\, R_1 = 10\,\text{k}\Omega$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3} \cdot \frac{1+sR_3C_2}{1+sR_4C_3}$ to the factors:

$$R_4 = 1\, R_3 = 10\,\text{k}\Omega, \qquad C_2 = \frac{1}{12600\, R_3} = 7.96\,\text{nF}, \qquad C_3 = \frac{1}{1.26 \times 10^5\, R_4} = 796\,\text{pF}$$

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 1 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.2\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 100}$	0.995	0.0997
$\left(1+\dfrac{s}{2\pi\cdot 2\text{k}}\right)$	1.118	0.464
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	0.9988	-0.05
total	**1.111** (product)	**0.513** (sum)

$$|H(j\omega_{in})| = 1.11, \qquad \angle H(j\omega_{in}) = 0.513 \text{ rad}$$
$$v_{o,ac}(t) = 1.11 \times 0.2\,\sin(2\pi\cdot 1000\,t + 0.513) = 0.222\,\sin(2\pi\cdot 1000\,t + 0.513)$$

$$\boxed{V_o(t) = 0.222\,\sin(2\pi\cdot 1000\,t + 0.513)\ \text{V}}$$

Answer

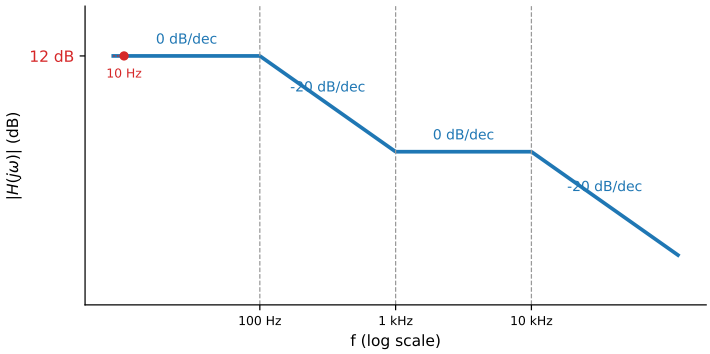
$$H(s) = 1\,\frac{s}{s+2\pi\cdot 100}\left(1+\frac{s}{2\pi\cdot 2\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

U1: R1 = 10kΩ, C1 = 159nF, R2 = 10kΩ

U2: R3 = 10kΩ, C2 = 7.96nF, R4 = 10kΩ, C3 = 796pF

$V_o(t) = 0.222 \sin(2\pi \cdot 1000\, t + 0.513)$ V

Solution 2C.10 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	0	-20	-20	one pole at f_c
1 kHz	-20	0	$+20$	one zero at f_c
10 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 12 dB at $f = 10$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 10$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	100 Hz	$f < f_c$: ≈ 1	0
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	10 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 12 - (0) = 12 \text{ dB} \quad \Rightarrow \quad K = 10^{12/20} = 3.98$$

So

$$\boxed{H(s) = 3.98\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	3.981	$\dfrac{1+\frac{s}{2\pi\cdot 1\text{k}}}{1+\frac{s}{2\pi\cdot 100}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$
U2	1	$\dfrac{1}{1+\frac{s}{2\pi\cdot 10\text{k}}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{1+sR_1C_1}{1+sR_2C_2}$ to the factors:

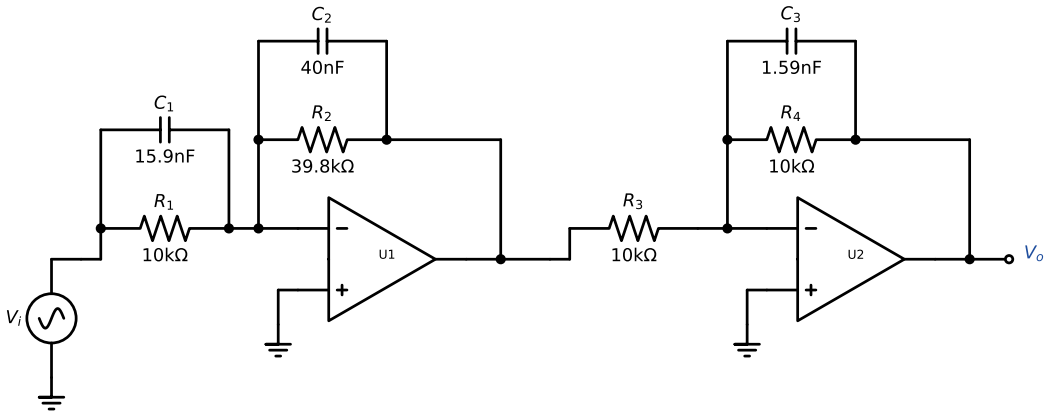
$$R_2 = 3.981\,R_1 = 39.8\,\text{k}\Omega, \qquad C_1 = \frac{1}{6280\,R_1} = 15.9\,\text{nF}, \qquad C_2 = \frac{1}{628\,R_2} = 40\,\text{nF}$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3}\cdot\frac{1}{1+sR_4C_3}$ to the factors:

$$R_4 = 1\,R_3 = 10\,\text{k}\Omega, \qquad C_3 = \frac{1}{62800\,R_4} = 1.59\,\text{nF}$$

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 3.981 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.2\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
3.98	3.981	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 100}}$	0.04994	-1.52
$\left(1+\dfrac{s}{2\pi \cdot 1\text{k}}\right)$	2.236	1.11
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 10\text{k}}}$	0.9806	-0.197
total	**0.4359** (product)	$-$**0.611** (sum)

$$|H(j\omega_{in})| = 0.436, \qquad \angle H(j\omega_{in}) = -0.611 \text{ rad}$$
$$v_{o,ac}(t) = 0.436 \times 0.2 \sin(2\pi \cdot 2000\,t - 0.611) = 0.0872 \sin(2\pi \cdot 2000\,t - 0.611)$$

$$\boxed{V_o(t) = 0.0872 \sin(2\pi \cdot 2000\,t - 0.611)\ \text{V}}$$

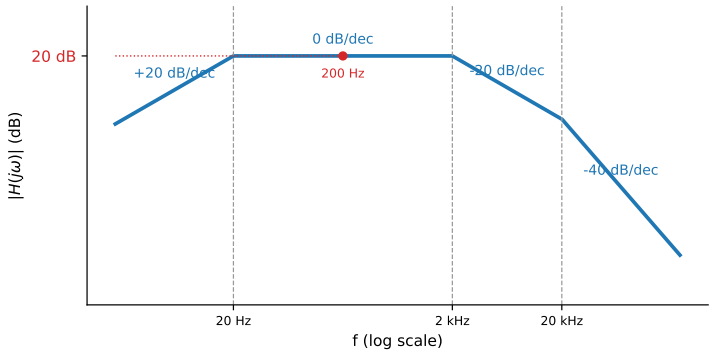
Answer

$$H(s) = 3.98\,\frac{1}{1+\dfrac{s}{2\pi \cdot 100}}\left(1+\frac{s}{2\pi \cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi \cdot 10\text{k}}}$$

U1: R1 = 10kΩ, C1 = 15.9nF, R2 = 39.8kΩ, C2 = 40nF
U2: R3 = 10kΩ, R4 = 10kΩ, C3 = 1.59nF

$V_o(t) = 0.0872 \sin(2\pi \cdot 2000\, t - 0.611)$ V

Solution 2C.11 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
20 Hz	$+20$	0	-20	one pole at f_c
2 kHz	0	-20	-20	one pole at f_c
20 kHz	-20	-40	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (20 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 20}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 20 dB at $f = 200$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 200$ Hz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 20}$	20 Hz	$f > f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 2\text{k}}}$	2 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}$	20 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20 \log_{10} K = 20 - (0) = 20 \text{ dB} \quad \Rightarrow \quad K = 10^{20/20} = 10$$

So

$$\boxed{H(s) = 10 \, \frac{s}{s + 2\pi \cdot 20} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 2\text{k}}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	10	$\dfrac{s}{s + 2\pi \cdot 20} \cdot \dfrac{1}{1 + \frac{s}{2\pi \cdot 2\text{k}}}$	band-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R \parallel C$
U2	1	$\dfrac{1}{1 + \frac{s}{2\pi \cdot 20\text{k}}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1} \cdot \frac{s}{s+1/(R_1 C_1)} \cdot \frac{1}{1+sR_2C_2}$ to the factors:

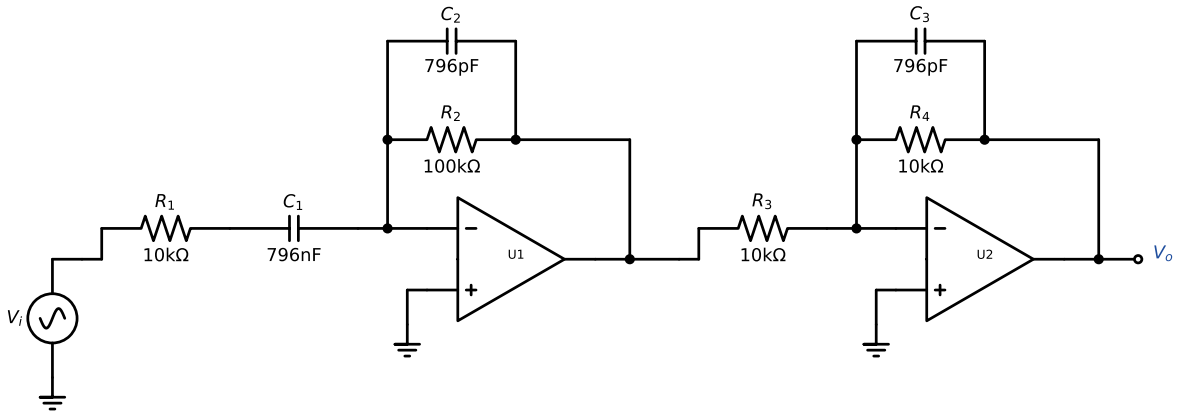
$$C_1 = \frac{1}{126\, R_1} = 796\,\text{nF}, \qquad R_2 = 10\, R_1 = 100\,\text{k}\Omega, \qquad C_2 = \frac{1}{12600\, R_2} = 796\,\text{pF}$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3} \cdot \frac{1}{1+sR_4C_3}$ to the factors:

$$R_4 = 1\, R_3 = 10\,\text{k}\Omega, \qquad C_3 = \frac{1}{1.26 \times 10^5\, R_4} = 796\,\text{pF}$$

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 10 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.1 \sin(2\pi \cdot 5000\, t)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	10	0
$\dfrac{s}{s+2\pi\cdot 20}$	1	0.004
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}$	0.3714	-1.19
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	0.9701	-0.245
total	**3.603** (product)	**−1.43** (sum)

$$|H(j\omega_{in})| = 3.6, \qquad \angle H(j\omega_{in}) = -1.43 \text{ rad}$$
$$v_{o,ac}(t) = 3.6 \times 0.1 \sin(2\pi \cdot 5000\, t - 1.43) = 0.36 \sin(2\pi \cdot 5000\, t - 1.43)$$

$$\boxed{V_o(t) = 0.36 \sin(2\pi \cdot 5000\, t - 1.43)\ \text{V}}$$

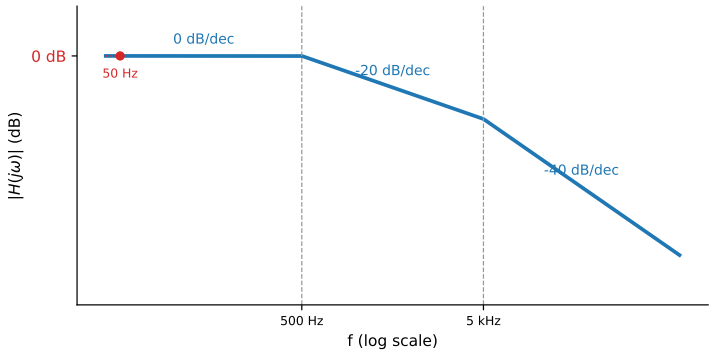
Answer

$$H(s) = 10\,\frac{s}{s+2\pi\cdot 20}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 2\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

U1: R1 = 10kΩ, C1 = 796nF, R2 = 100kΩ, C2 = 796pF
U2: R3 = 10kΩ, R4 = 10kΩ, C3 = 796pF

$V_o(t) = 0.36 \sin(2\pi \cdot 5000\, t - 1.43)$ V

Solution 2C.12 **Medium**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
500 Hz	0	-20	-20	one pole at f_c
5 kHz	-20	-40	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 500}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 0 dB at $f = 50$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 50$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 500}}$	500 Hz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}$	5 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 0 - (0) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi\cdot 500}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	1	$\dfrac{1}{1+\frac{s}{2\pi\cdot 500}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$
U2	1	$\dfrac{1}{1+\frac{s}{2\pi\cdot 5\text{k}}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{1}{1+sR_2C_1}$ to the factors:

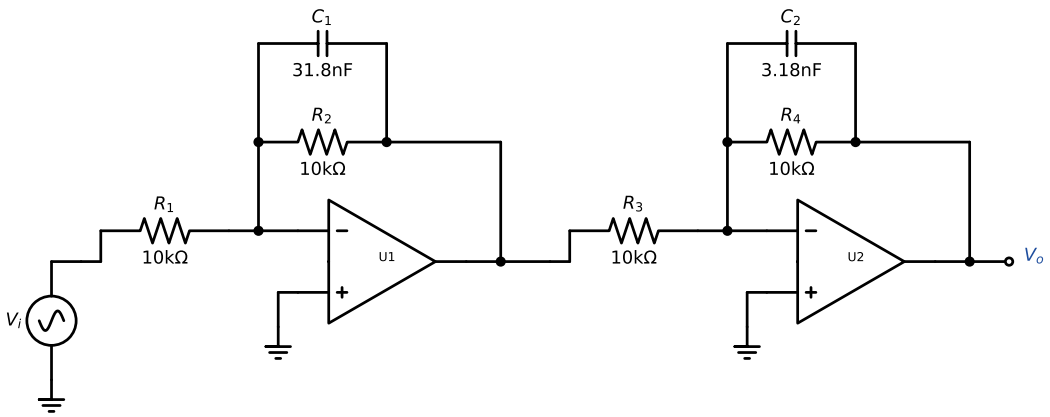
$$R_2 = 1\,R_1 = 10\,\text{k}\Omega, \qquad C_1 = \frac{1}{3140\,R_2} = 31.8\,\text{nF}$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3}\cdot\frac{1}{1+sR_4C_2}$ to the factors:

$$R_4 = 1\,R_3 = 10\,\text{k}\Omega, \qquad C_2 = \frac{1}{31400\,R_4} = 3.18\,\text{nF}$$

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 1 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 1 \sin\left(2\pi \cdot 1000\, t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 500}}$	0.4472	-1.11
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 5\text{k}}}$	0.9806	-0.197
total	**0.4385** (product)	$\mathbf{-1.3}$ (sum)

$$|H(j\omega_{in})| = 0.439, \qquad \angle H(j\omega_{in}) = -1.3 \text{ rad}$$
$$v_{o,ac}(t) = 0.439 \times 1 \sin\left(2\pi \cdot 1000\, t - 0.519\right) = 0.439 \sin\left(2\pi \cdot 1000\, t - 0.519\right)$$

(phase: $\theta + \angle H = \frac{\pi}{4} + (-1.3) = -0.519\,\text{rad}$.)

$$\boxed{V_o(t) = 0.439 \sin\left(2\pi \cdot 1000\, t - 0.519\right)\ \text{V}}$$

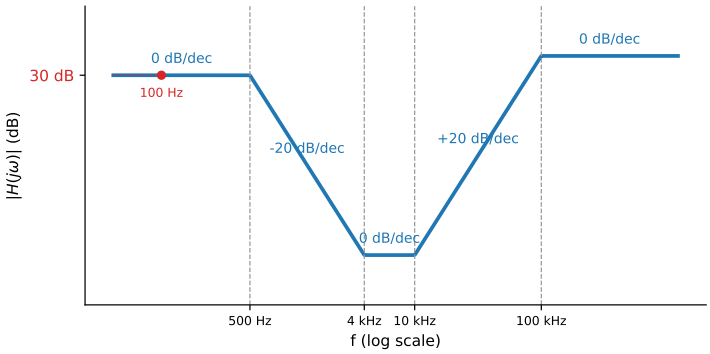
Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi \cdot 500}}\,\frac{1}{1+\dfrac{s}{2\pi \cdot 5\text{k}}}$$

U1: R1 = 10kΩ, R2 = 10kΩ, C1 = 31.8nF
U2: R3 = 10kΩ, R4 = 10kΩ, C2 = 3.18nF

$V_o(t) = 0.439 \sin\left(2\pi \cdot 1000\, t - 0.519\right)$ V

Solution 2C.13 **Hard**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
500 Hz	0	-20	-20	one pole at f_c
4 kHz	-20	0	$+20$	one zero at f_c
10 kHz	0	$+20$	$+20$	one zero at f_c
100 kHz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 500}}\left(1+\frac{s}{2\pi\cdot 4\text{k}}\right)\left(1+\frac{s}{2\pi\cdot 10\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 30 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 500}}$	500 Hz	$f < f_c$: ≈ 1	0
$\left(1+\dfrac{s}{2\pi\cdot 4\text{k}}\right)$	4 kHz	$f < f_c$: ≈ 1	0
$\left(1+\dfrac{s}{2\pi\cdot 10\text{k}}\right)$	10 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}$	100 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 30 - (0) = 30 \text{ dB} \quad \Rightarrow \quad K = 10^{30/20} = 31.6$$

So

$$\boxed{H(s) = 31.6\,\frac{1}{1+\dfrac{s}{2\pi\cdot 500}}\left(1+\frac{s}{2\pi\cdot 4\text{k}}\right)\left(1+\frac{s}{2\pi\cdot 10\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	31.62	$\dfrac{1+\frac{s}{2\pi\cdot 4\text{k}}}{1+\frac{s}{2\pi\cdot 500}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$
U2	1	$\dfrac{1+\frac{s}{2\pi\cdot 10\text{k}}}{1+\frac{s}{2\pi\cdot 100\text{k}}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{1+sR_1C_1}{1+sR_2C_2}$ to the factors:

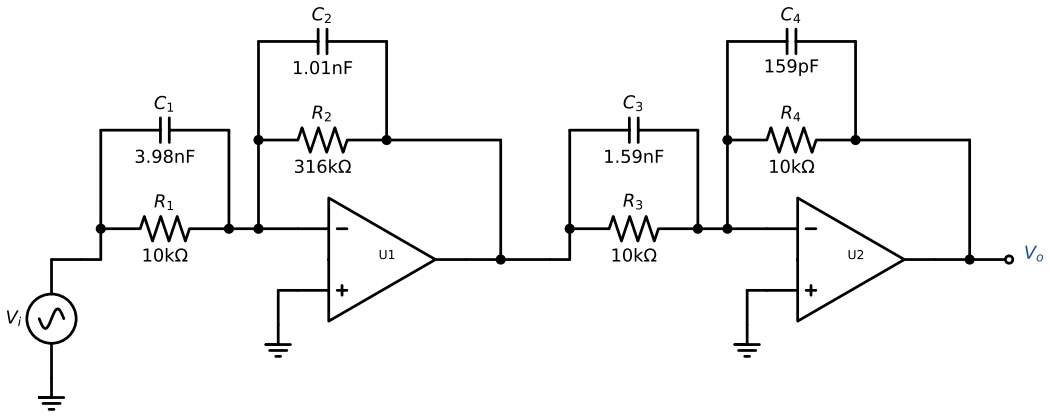
$$R_2 = 31.62\,R_1 = 316\,\text{k}\Omega, \qquad C_1 = \frac{1}{25100\,R_1} = 3.98\,\text{nF}, \qquad C_2 = \frac{1}{3140\,R_2} = 1.01\,\text{nF}$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3}\cdot\frac{1+sR_3C_3}{1+sR_4C_4}$ to the factors:

$$R_4 = 1\,R_3 = 10\,\text{k}\Omega, \qquad C_3 = \frac{1}{62800\,R_3} = 1.59\,\text{nF}, \qquad C_4 = \frac{1}{6.28\times 10^5\,R_4} = 159\,\text{pF}$$

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 31.62 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.01 \sin(2\pi \cdot 2000\, t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
31.6	31.62	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 500}}$	0.2425	-1.33
$\left(1+\dfrac{s}{2\pi \cdot 4\text{k}}\right)$	1.118	0.464
$\left(1+\dfrac{s}{2\pi \cdot 10\text{k}}\right)$	1.02	0.197
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 100\text{k}}}$	0.9998	-0.02
total	**8.743** (product)	$-$**0.685** (sum)

$$|H(j\omega_{in})| = 8.74, \qquad \angle H(j\omega_{in}) = -0.685 \text{ rad}$$
$$v_{o,ac}(t) = 8.74 \times 0.01 \sin(2\pi \cdot 2000\, t - 0.685) = 0.0874 \sin(2\pi \cdot 2000\, t - 0.685)$$

$$\boxed{V_o(t) = 0.0874 \sin(2\pi \cdot 2000\, t - 0.685)\ \text{V}}$$

Answer

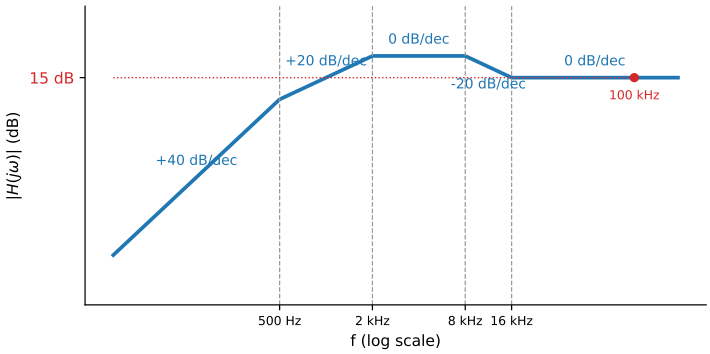
$$H(s) = 31.6\,\frac{1}{1+\dfrac{s}{2\pi\cdot 500}}\left(1+\frac{s}{2\pi\cdot 4\text{k}}\right)\left(1+\frac{s}{2\pi\cdot 10\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}$$

U1: R1 = 10kΩ, C1 = 3.98nF, R2 = 316kΩ, C2 = 1.01nF

U2: R3 = 10kΩ, C3 = 1.59nF, R4 = 10kΩ, C4 = 159pF

$$V_o(t) = 0.0874\,\sin\left(2\pi\cdot 2000\,t - 0.685\right)\ \text{V}$$

Solution 2C.14 — Hard



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
500 Hz	$+40$	$+20$	-20	one pole at f_c
2 kHz	$+20$	0	-20	one pole at f_c
8 kHz	0	-20	-20	one pole at f_c
16 kHz	-20	0	$+20$	one zero at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+40$ dB/dec, i.e. $|H| \propto \omega^2$ at low frequency: the numerator contains s^2. As in class, pair each s with one of the *lowest* poles (500 Hz, 2 kHz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 500}\,\frac{s}{s+2\pi\cdot 2\text{k}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 8\text{k}}}\left(1+\frac{s}{2\pi\cdot 16\text{k}}\right)$$

Step 4 — Find K from the marked level

The plot marks the level 15 dB at $f = 100$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ kHz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 500}$	500 Hz	$f > f_c$: ≈ 1	0
$\dfrac{s}{s + 2\pi \cdot 2\text{k}}$	2 kHz	$f > f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 8\text{k}}}$	8 kHz	$f > f_c$: $\approx \frac{p}{\omega}$	-21.9
$\left(1 + \dfrac{s}{2\pi \cdot 16\text{k}}\right)$	16 kHz	$f > f_c$: $\approx \frac{\omega}{z}$	15.9
sum			$\mathbf{-6.02}$

The dB values of the factors *add* (magnitudes multiply), so

$$20 \log_{10} K = 15 - (-6.02) = 21 \text{ dB} \quad \Rightarrow \quad K = 10^{21/20} = 11.2$$

So

$$\boxed{H(s) = 11.2 \, \frac{s}{s + 2\pi \cdot 500} \, \frac{s}{s + 2\pi \cdot 2\text{k}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 8\text{k}}} \left(1 + \frac{s}{2\pi \cdot 16\text{k}}\right)}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	11.25	$\dfrac{s}{s + 2\pi \cdot 500}$	high-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R$
U2	1	$\dfrac{s}{s + 2\pi \cdot 2\text{k}}$	high-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R$
U3	1	$\dfrac{1 + \frac{s}{2\pi \cdot 16\text{k}}}{1 + \frac{s}{2\pi \cdot 8\text{k}}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$
U4	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1} \cdot \frac{s}{s+1/(R_1 C_1)}$ to the factors:

$$C_1 = \frac{1}{3140\, R_1} = 31.8\,\text{nF}, \qquad R_2 = 11.25\, R_1 = 112\,\text{k}\Omega$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3}\cdot\frac{s}{s+1/(R_3C_2)}$ to the factors:

$$C_2 = \frac{1}{12600\,R_3} = 7.96\,\text{nF}, \qquad R_4 = 1\,R_3 = 10\,\text{k}\Omega$$

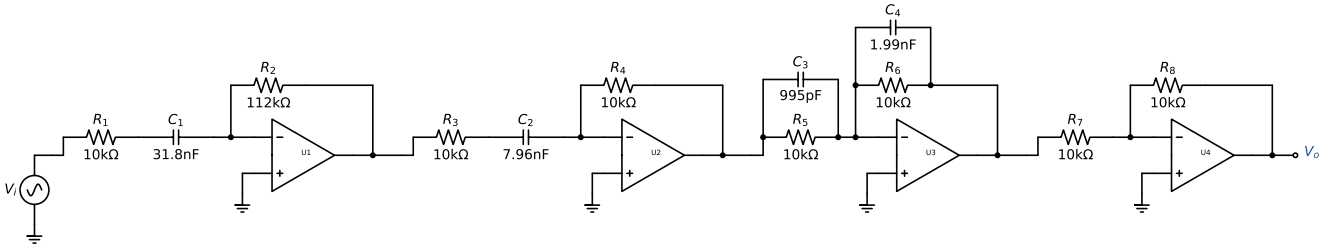
3. U3: choose $R_5 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_6}{R_5}\cdot\frac{1+sR_5C_3}{1+sR_6C_4}$ to the factors:

$$R_6 = 1\,R_5 = 10\,\text{k}\Omega, \qquad C_3 = \frac{1}{1.01\times10^5\,R_5} = 995\,\text{pF}, \qquad C_4 = \frac{1}{50300\,R_6} = 1.99\,\text{nF}$$

4. U4: $R_7 = R_8 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^4\cdot 11.25\cdot(\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.3\sin(2\pi\cdot 10000\,t)$, so $\omega_{in} = 2\pi\cdot 10000 = 62800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
11.2	11.25	0
$\dfrac{s}{s+2\pi\cdot 500}$	0.9988	0.05
$\dfrac{s}{s+2\pi\cdot 2\text{k}}$	0.9806	0.197
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 8\text{k}}}$	0.6247	−0.896
$\left(1+\dfrac{s}{2\pi\cdot 16\text{k}}\right)$	1.179	0.559
total	**8.114** (product)	**−0.0901** (sum)

$$|H(j\omega_{in})| = 8.11, \qquad \angle H(j\omega_{in}) = -0.0901\ \text{rad}$$
$$v_{o,ac}(t) = 8.11\times 0.3\,\sin(2\pi\cdot 10000\,t - 0.0901) = 2.43\,\sin(2\pi\cdot 10000\,t - 0.0901)$$

$$\boxed{V_o(t) = 2.43\,\sin(2\pi\cdot 10000\,t - 0.0901)\ \text{V}}$$

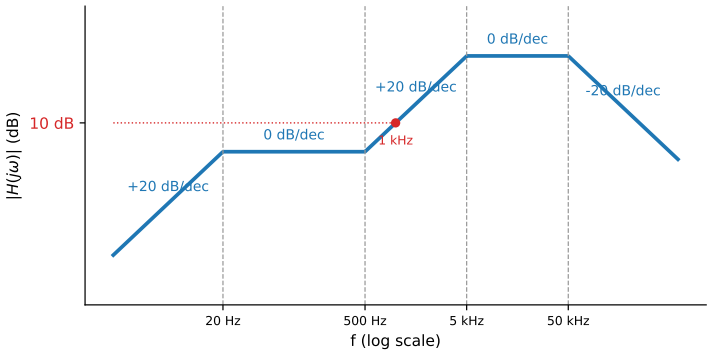
Answer

$$H(s) = 11.2\,\frac{s}{s+2\pi\cdot 500}\,\frac{s}{s+2\pi\cdot 2\text{k}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 8\text{k}}}\left(1+\frac{s}{2\pi\cdot 16\text{k}}\right)$$

U1: R1 = 10kΩ, C1 = 31.8nF, R2 = 112kΩ
U2: R3 = 10kΩ, C2 = 7.96nF, R4 = 10kΩ
U3: R5 = 10kΩ, C3 = 995pF, R6 = 10kΩ, C4 = 1.99nF
U4: R7 = 10kΩ, R8 = 10kΩ

$$V_o(t) = 2.43\,\sin\left(2\pi\cdot 10000\,t - 0.0901\right)\ \text{V}$$

Solution 2C.15 **Hard**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
20 Hz	$+20$	0	-20	one pole at f_c
500 Hz	0	$+20$	$+20$	one zero at f_c
5 kHz	$+20$	0	-20	one pole at f_c
50 kHz	0	-20	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (20 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 20}\left(1+\frac{s}{2\pi\cdot 500}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 5\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 50\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 10 dB at $f = 1$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 1$ kHz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 20}$	20 Hz	$f > f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 500}\right)$	500 Hz	$f > f_c$: $\approx \frac{\omega}{z}$	6.02
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}}$	5 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}$	50 kHz	$f < f_c$: ≈ 1	0
sum			**6.02**

The dB values of the factors *add* (magnitudes multiply), so

$$20 \log_{10} K = 10 - (6.02) = 3.98 \text{ dB} \quad \Rightarrow \quad K = 10^{3.98/20} = 1.58$$

So

$$\boxed{H(s) = 1.58 \, \frac{s}{s + 2\pi \cdot 20} \left(1 + \frac{s}{2\pi \cdot 500}\right) \frac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	1.581	$\dfrac{s}{s + 2\pi \cdot 20} \cdot \dfrac{1}{1 + \frac{s}{2\pi \cdot 50\text{k}}}$	band-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R \parallel C$
U2	1	$\dfrac{1 + \frac{s}{2\pi \cdot 500}}{1 + \frac{s}{2\pi \cdot 5\text{k}}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1} \cdot \frac{s}{s+1/(R_1 C_1)} \cdot \frac{1}{1+sR_2C_2}$ to the factors:

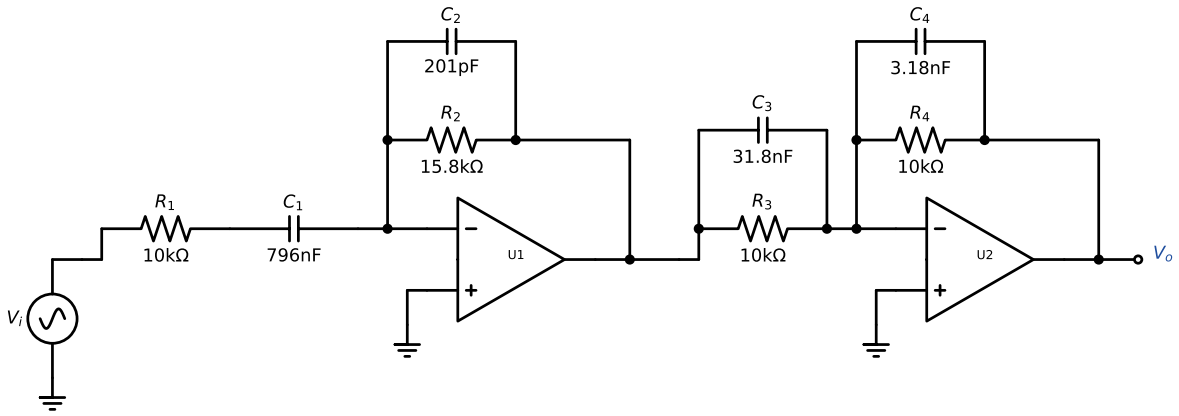
$$C_1 = \frac{1}{126\, R_1} = 796\,\text{nF}, \qquad R_2 = 1.581\, R_1 = 15.8\,\text{k}\Omega, \qquad C_2 = \frac{1}{3.14 \times 10^5\, R_2} = 201\,\text{pF}$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3} \cdot \frac{1+sR_3C_3}{1+sR_4C_4}$ to the factors:

$$R_4 = 1\, R_3 = 10\,\text{k}\Omega, \qquad C_3 = \frac{1}{3140\, R_3} = 31.8\,\text{nF}, \qquad C_4 = \frac{1}{31400\, R_4} = 3.18\,\text{nF}$$

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^2 \cdot 1.581 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.2 \sin\left(2\pi \cdot 2000\, t + \frac{\pi}{3}\right)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1.58	1.581	0
$\dfrac{s}{s + 2\pi \cdot 20}$	1	0.01
$\left(1 + \dfrac{s}{2\pi \cdot 500}\right)$	4.123	1.33
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}}$	0.9285	−0.381
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}$	0.9992	−0.04
total	**6.048** (product)	**0.915** (sum)

$$|H(j\omega_{in})| = 6.05, \qquad \angle H(j\omega_{in}) = 0.915 \text{ rad}$$
$$v_{o,ac}(t) = 6.05 \times 0.2 \sin\left(2\pi \cdot 2000\, t + 1.96\right) = 1.21 \sin\left(2\pi \cdot 2000\, t + 1.96\right)$$

(phase: $\theta + \angle H = \frac{\pi}{3} + (0.915) = 1.96$ rad.)

$$\boxed{V_o(t) = 1.21 \sin\left(2\pi \cdot 2000\, t + 1.96\right) \text{ V}}$$

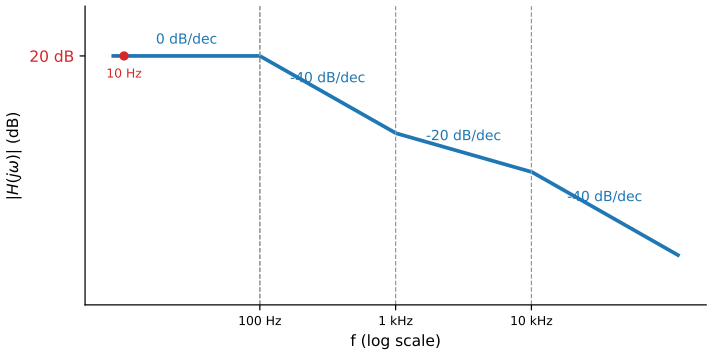
Answer

$$H(s) = 1.58\,\frac{s}{s + 2\pi \cdot 20}\left(1 + \frac{s}{2\pi \cdot 500}\right)\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 5\text{k}}}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}$$

U1: R1 = 10kΩ, C1 = 796nF, R2 = 15.8kΩ, C2 = 201pF
U2: R3 = 10kΩ, C3 = 31.8nF, R4 = 10kΩ, C4 = 3.18nF

$$V_o(t) = 1.21\,\sin\left(2\pi \cdot 2000\,t + 1.96\right)\ \text{V}$$

Solution 2C.16 **Hard**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
100 Hz	0	-40	-40	two poles at f_c
1 kHz	-40	-20	$+20$	one zero at f_c
10 kHz	-20	-40	-20	one pole at f_c

Step 2 — Starting slope $\Rightarrow$ factors of s

The plot starts *flat* (0 dB/dec), so there is no factor s in the numerator: every pole is written as $\frac{1}{1+s/p}$ and every zero as $\left(1+\frac{s}{z}\right)$, which are all ≈ 1 (0 dB) at low frequency. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 20 dB at $f = 10$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 10$ Hz	asymptote (dB)
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	100 Hz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	100 Hz	$f < f_c$: ≈ 1	0
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	10 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20\log_{10} K = 20 - (0) = 20 \text{ dB} \quad \Rightarrow \quad K = 10^{20/20} = 10$$

So

$$\boxed{H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\,\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\,\frac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	10	$\dfrac{1+\frac{s}{2\pi\cdot 1\text{k}}}{1+\frac{s}{2\pi\cdot 100}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$
U2	1	$\dfrac{1}{1+\frac{s}{2\pi\cdot 100}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$
U3	1	$\dfrac{1}{1+\frac{s}{2\pi\cdot 10\text{k}}}$	low-pass: $Z_{in} = R$, $Z_f = R \parallel C$
U4	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1}\cdot\frac{1+sR_1C_1}{1+sR_2C_2}$ to the factors:

$$R_2 = 10\,R_1 = 100\,\text{k}\Omega, \qquad C_1 = \frac{1}{6280\,R_1} = 15.9\,\text{nF}, \qquad C_2 = \frac{1}{628\,R_2} = 15.9\,\text{nF}$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3}\cdot\frac{1}{1+sR_4C_3}$ to the factors:

$$R_4 = 1\,R_3 = 10\,\text{k}\Omega, \qquad C_3 = \frac{1}{628\,R_4} = 159\,\text{nF}$$

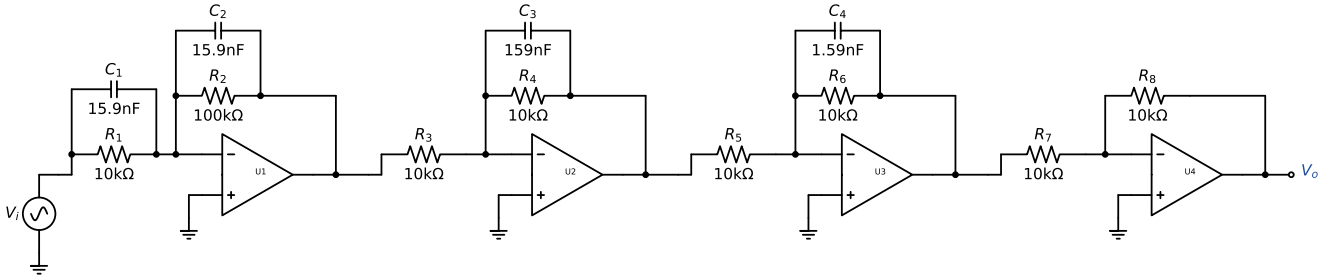
3. U3: choose $R_5 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_6}{R_5}\cdot\frac{1}{1+sR_6C_4}$ to the factors:

$$R_6 = 1\,R_5 = 10\,\text{k}\Omega, \qquad C_4 = \frac{1}{62800\,R_6} = 1.59\,\text{nF}$$

4. U4: $R_7 = R_8 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^4\cdot 10\cdot(\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.1\sin(2\pi\cdot 300\,t)$, so $\omega_{in} = 2\pi\cdot 300 = 1880\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	10	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	0.3162	-1.25
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	0.3162	-1.25
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	1.044	0.291
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 10\text{k}}}$	0.9996	-0.03
total	**1.044** (product)	$-$**2.24** (sum)

$$|H(j\omega_{in})| = 1.04, \qquad \angle H(j\omega_{in}) = -2.24 \text{ rad}$$
$$v_{o,ac}(t) = 1.04 \times 0.1 \sin(2\pi \cdot 300\,t - 2.24) = 0.104 \sin(2\pi \cdot 300\,t - 2.24)$$

$$\boxed{V_o(t) = 0.104 \sin(2\pi \cdot 300\,t - 2.24) \text{ V}}$$

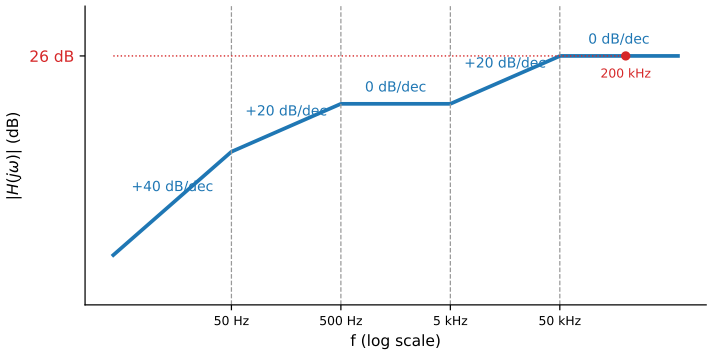
Answer

$$H(s) = 10 \frac{1}{1 + \frac{s}{2\pi \cdot 100}} \frac{1}{1 + \frac{s}{2\pi \cdot 100}} \left(1 + \frac{s}{2\pi \cdot 1\text{k}}\right) \frac{1}{1 + \frac{s}{2\pi \cdot 10\text{k}}}$$

U1: R1 = 10kΩ, C1 = 15.9nF, R2 = 100kΩ, C2 = 15.9nF
U2: R3 = 10kΩ, R4 = 10kΩ, C3 = 159nF
U3: R5 = 10kΩ, R6 = 10kΩ, C4 = 1.59nF
U4: R7 = 10kΩ, R8 = 10kΩ

$V_o(t) = 0.104 \sin(2\pi \cdot 300\,t - 2.24)$ V

Solution 2C.17 Hard



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
50 Hz	$+40$	$+20$	-20	one pole at f_c
500 Hz	$+20$	0	-20	one pole at f_c
5 kHz	0	$+20$	$+20$	one zero at f_c
50 kHz	$+20$	0	-20	one pole at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+40$ dB/dec, i.e. $|H| \propto \omega^2$ at low frequency: the numerator contains s^2. As in class, pair each s with one of the *lowest* poles (50 Hz, 500 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 50}\,\frac{s}{s+2\pi\cdot 500}\left(1+\frac{s}{2\pi\cdot 5\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 50\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 26 dB at $f = 200$ kHz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 200$ kHz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 50}$	50 Hz	$f > f_c$: ≈ 1	0
$\dfrac{s}{s + 2\pi \cdot 500}$	500 Hz	$f > f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 5\text{k}}\right)$	5 kHz	$f > f_c$: $\approx \frac{\omega}{z}$	32
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}$	50 kHz	$f > f_c$: $\approx \frac{p}{\omega}$	-12
sum			**20**

The dB values of the factors *add* (magnitudes multiply), so

$$20 \log_{10} K = 26 - (20) = 6 \text{ dB} \quad \Rightarrow \quad K = 10^{6/20} = 2$$

So

$$\boxed{H(s) = 2 \, \frac{s}{s + 2\pi \cdot 50} \, \frac{s}{s + 2\pi \cdot 500} \left(1 + \frac{s}{2\pi \cdot 5\text{k}}\right) \frac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	1.995	$\dfrac{s}{s + 2\pi \cdot 50}$	high-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R$
U2	1	$\dfrac{s}{s + 2\pi \cdot 500}$	high-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R$
U3	1	$\dfrac{1 + \frac{s}{2\pi \cdot 5\text{k}}}{1 + \frac{s}{2\pi \cdot 50\text{k}}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$
U4	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1} \cdot \frac{s}{s+1/(R_1 C_1)}$ to the factors:

$$C_1 = \frac{1}{314\, R_1} = 318\,\text{nF}, \qquad R_2 = 1.995\, R_1 = 20\,\text{k}\Omega$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3}\cdot\frac{s}{s+1/(R_3C_2)}$ to the factors:

$$C_2 = \frac{1}{3140\,R_3} = 31.8\,\text{nF}, \qquad R_4 = 1\,R_3 = 10\,\text{k}\Omega$$

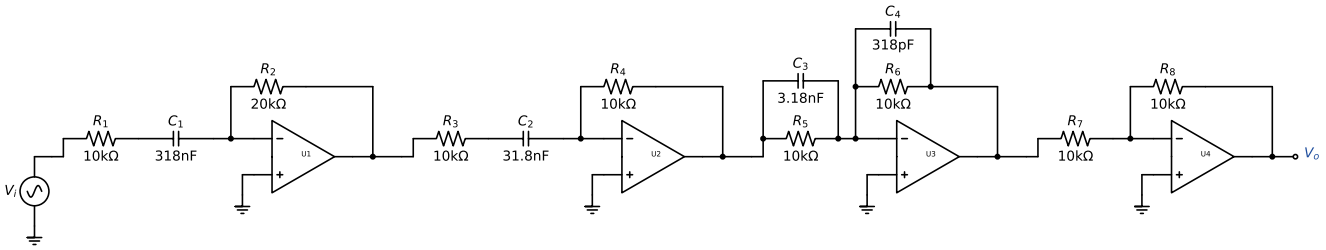
3. U3: choose $R_5 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_6}{R_5}\cdot\frac{1+sR_5C_3}{1+sR_6C_4}$ to the factors:

$$R_6 = 1\,R_5 = 10\,\text{k}\Omega, \qquad C_3 = \frac{1}{31400\,R_5} = 3.18\,\text{nF}, \qquad C_4 = \frac{1}{3.14\times10^5\,R_6} = 318\,\text{pF}$$

4. U4: $R_7 = R_8 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^4\cdot 1.995\cdot(\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.05\cos\left(2\pi\cdot1000\,t - \frac{\pi}{6}\right)$, so $\omega_{in} = 2\pi\cdot1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
2	1.995	0
$\dfrac{s}{s+2\pi\cdot50}$	0.9988	0.05
$\dfrac{s}{s+2\pi\cdot500}$	0.8944	0.464
$\left(1+\dfrac{s}{2\pi\cdot5\text{k}}\right)$	1.02	0.197
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot50\text{k}}}$	0.9998	-0.02
total	**1.817** (product)	**0.691** (sum)

$$|H(j\omega_{in})| = 1.82, \qquad \angle H(j\omega_{in}) = 0.691\ \text{rad}$$
$$v_{o,ac}(t) = 1.82\times0.05\,\cos\left(2\pi\cdot1000\,t + 0.167\right) = 0.0909\,\cos\left(2\pi\cdot1000\,t + 0.167\right)$$

(phase: $\theta + \angle H = -\frac{\pi}{6} + (0.691) = 0.167\,\text{rad.}$)

$$\boxed{V_o(t) = 0.0909\,\cos\left(2\pi \cdot 1000\,t + 0.167\right)\ \text{V}}$$

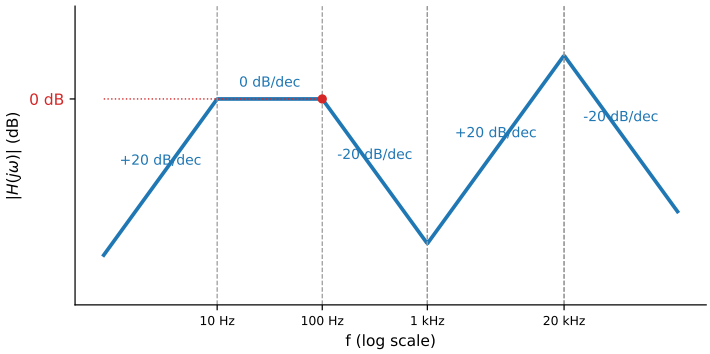
Answer

$$H(s) = 2\,\frac{s}{s + 2\pi \cdot 50}\,\frac{s}{s + 2\pi \cdot 500}\left(1 + \frac{s}{2\pi \cdot 5\text{k}}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 50\text{k}}}$$

U1: R1 = 10kΩ, C1 = 318nF, R2 = 20kΩ
U2: R3 = 10kΩ, C2 = 31.8nF, R4 = 10kΩ
U3: R5 = 10kΩ, C3 = 3.18nF, R6 = 10kΩ, C4 = 318pF
U4: R7 = 10kΩ, R8 = 10kΩ

$V_o(t) = 0.0909\,\cos\left(2\pi \cdot 1000\,t + 0.167\right)\ \text{V}$

Solution 2C.18 **Hard**



Step 1 — Read the corners and slopes

Go left to right along the plot and write down the slope of every straight segment. At each corner frequency the slope changes by a multiple of 20 dB/dec: every -20 is one pole, every $+20$ is one zero (tool T6 read backwards).

corner f_c	slope before	slope after	change	meaning
10 Hz	$+20$	0	-20	one pole at f_c
100 Hz	0	-20	-20	one pole at f_c
1 kHz	-20	$+20$	$+40$	two zeros at f_c
20 kHz	$+20$	-20	-40	two poles at f_c

Step 2 — Starting slope ⇒ factors of s

The plot starts with a slope of $+20$ dB/dec, i.e. $|H| \propto \omega^1$ at low frequency: the numerator contains s. As in class, pair each s with one of the *lowest* poles (10 Hz) as $\frac{s}{s+p}$: such a factor rises at $+20$ dB/dec below p and is ≈ 1 (0 dB) above it. The remaining poles are written $\frac{1}{1+s/p}$ and the zeros $\left(1+\frac{s}{z}\right)$. Every corner is an angular frequency $\omega_c = 2\pi f_c$.

Step 3 — Assemble $H(s)$

Put one factor per pole/zero in front of an unknown constant K:

$$H(s) = K\,\frac{s}{s+2\pi\cdot 10}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

Step 4 — Find K from the marked level

The plot marks the level 0 dB at $f = 100$ Hz. On the straight-line plot every factor is either exactly 0 dB (where it is ≈ 1) or a ± 20 dB/dec line, so evaluate the *asymptote* of each factor at that frequency (the level of a line with slope ± 20 dB/dec is $\pm 20\log_{10}(f/f_c)$):

factor	f_c	at $f = 100$ Hz	asymptote (dB)
$\dfrac{s}{s + 2\pi \cdot 10}$	10 Hz	$f > f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 100}}$	100 Hz	$f < f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
$\left(1 + \dfrac{s}{2\pi \cdot 1\text{k}}\right)$	1 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}$	20 kHz	$f < f_c$: ≈ 1	0
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}$	20 kHz	$f < f_c$: ≈ 1	0
sum			**0**

The dB values of the factors *add* (magnitudes multiply), so

$$20 \log_{10} K = 0 - (0) = 0 \text{ dB} \quad \Rightarrow \quad K = 10^{0/20} = 1$$

So

$$\boxed{H(s) = 1 \, \frac{s}{s + 2\pi \cdot 10} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 100}} \, \left(1 + \frac{s}{2\pi \cdot 1\text{k}}\right) \left(1 + \frac{s}{2\pi \cdot 1\text{k}}\right) \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 20\text{k}}}}$$

Step 5 — Split $H(s)$ into op-amp stages

Every inverting stage of the form $-\frac{Z_f}{Z_{in}}$ with Z_{in}, Z_f made of one R and at most one C realizes *one* pole, optionally with one zero or one s (tool T2). Pair each zero with a low-pass pole (pole-zero stage), each $\frac{s}{s+p}$ with a remaining low-pass pole (band-pass stage), and give every leftover factor its own stage. Put all of $|K|$ into the first stage. Each stage inverts, so with an odd number of stages add a unity-gain inverter to make $K > 0$.

stage	gain $\frac{R_2}{R_1}$	factor(s) realized	block
U1	1	$\dfrac{s}{s + 2\pi \cdot 10} \cdot \dfrac{1}{1 + \frac{s}{2\pi \cdot 20\text{k}}}$	band-pass: $Z_{in} = R + \frac{1}{sC}$, $Z_f = R \parallel C$
U2	1	$\dfrac{1 + \frac{s}{2\pi \cdot 1\text{k}}}{1 + \frac{s}{2\pi \cdot 100}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$
U3	1	$\dfrac{1 + \frac{s}{2\pi \cdot 1\text{k}}}{1 + \frac{s}{2\pi \cdot 20\text{k}}}$	pole-zero: $Z_{in} = R \parallel C$, $Z_f = R \parallel C$
U4	1	1	unity inverter (fixes the sign)

Step 6 — Component values

Use $R_1 = 10\,\text{k}\Omega$ at the input of every stage (any convenient value works; it only scales the other parts). The corner of a factor $1 + s/\omega_c$ is matched by $RC = 1/\omega_c$ (corners in rad/s).

1. U1: choose $R_1 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_2}{R_1} \cdot \frac{s}{s+1/(R_1C_1)} \cdot \frac{1}{1+sR_2C_2}$ to the factors:

$$C_1 = \frac{1}{62.8\,R_1} = 1.59\,\mu\text{F}, \qquad R_2 = 1\,R_1 = 10\,\text{k}\Omega, \qquad C_2 = \frac{1}{1.26\times10^5\,R_2} = 796\,\text{pF}$$

2. U2: choose $R_3 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_4}{R_3} \cdot \frac{1+sR_3C_3}{1+sR_4C_4}$ to the factors:

$$R_4 = 1\,R_3 = 10\,\text{k}\Omega, \qquad C_3 = \frac{1}{6280\,R_3} = 15.9\,\text{nF}, \qquad C_4 = \frac{1}{628\,R_4} = 159\,\text{nF}$$

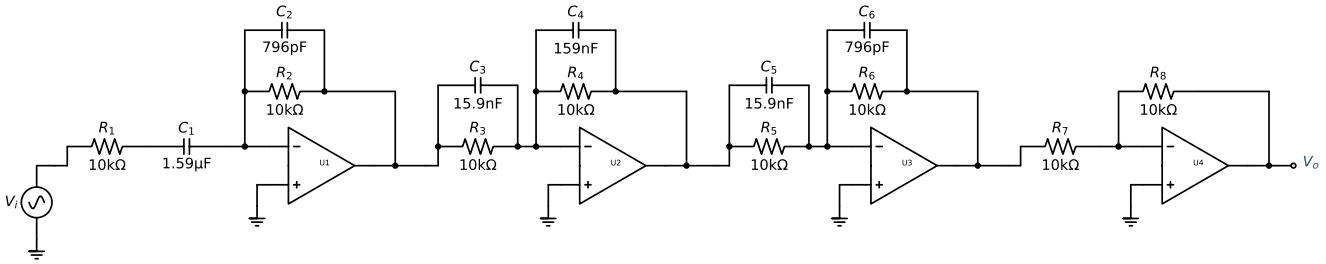
3. U3: choose $R_5 = 10\,\text{k}\Omega$, then match the stage formula $-\frac{R_6}{R_5} \cdot \frac{1+sR_5C_5}{1+sR_6C_6}$ to the factors:

$$R_6 = 1\,R_5 = 10\,\text{k}\Omega, \qquad C_5 = \frac{1}{6280\,R_5} = 15.9\,\text{nF}, \qquad C_6 = \frac{1}{1.26\times10^5\,R_6} = 796\,\text{pF}$$

4. U4: $R_7 = R_8 = 10\,\text{k}\Omega$ (gain -1).

Step 7 — Check and steady-state output

The designed circuit:



Check: the stage gains multiply to $(-1)^4 \cdot 1 \cdot (\text{factors}) = H(s)$; an independent nodal solve of this circuit reproduces $H(s)$ at every frequency. (Standard-value resistors and capacitors would move the corners slightly.)

The input is $V_i(t) = 0.5\sin(2\pi \cdot 3000\,t)$, so $\omega_{in} = 2\pi \cdot 3000 = 18800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 10}$	1	0.00333
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100}}$	0.03331	-1.54
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	3.162	1.25
$\left(1+\dfrac{s}{2\pi\cdot 1\text{k}}\right)$	3.162	1.25
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	0.9889	-0.149
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$	0.9889	-0.149
total	**0.3258** (product)	**0.666** (sum)

$$|H(j\omega_{in})| = 0.326, \qquad \angle H(j\omega_{in}) = 0.666 \text{ rad}$$
$$v_{o,ac}(t) = 0.326 \times 0.5 \sin\left(2\pi\cdot 3000\,t + 0.666\right) = 0.163 \sin\left(2\pi\cdot 3000\,t + 0.666\right)$$

$$\boxed{V_o(t) = 0.163 \sin\left(2\pi\cdot 3000\,t + 0.666\right)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 10}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 100}}\,\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\,\left(1+\frac{s}{2\pi\cdot 1\text{k}}\right)\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 20\text{k}}}$$

U1: R1 = 10kΩ, C1 = 1.59μF, R2 = 10kΩ, C2 = 796pF
U2: R3 = 10kΩ, C3 = 15.9nF, R4 = 10kΩ, C4 = 159nF
U3: R5 = 10kΩ, C5 = 15.9nF, R6 = 10kΩ, C6 = 796pF
U4: R7 = 10kΩ, R8 = 10kΩ

$V_o(t) = 0.163 \sin\left(2\pi\cdot 3000\,t + 0.666\right)$ V